

Offline Reinforcement Learning

Algorithms, Modern Instantiations, and Applications

Aviral Kumar

Carnegie Mellon University

School of Computer Science

Online RL: Inability to Collect Lots of Data

rely exclusively on interaction

Expense

Safety

~1-2 months /
data point



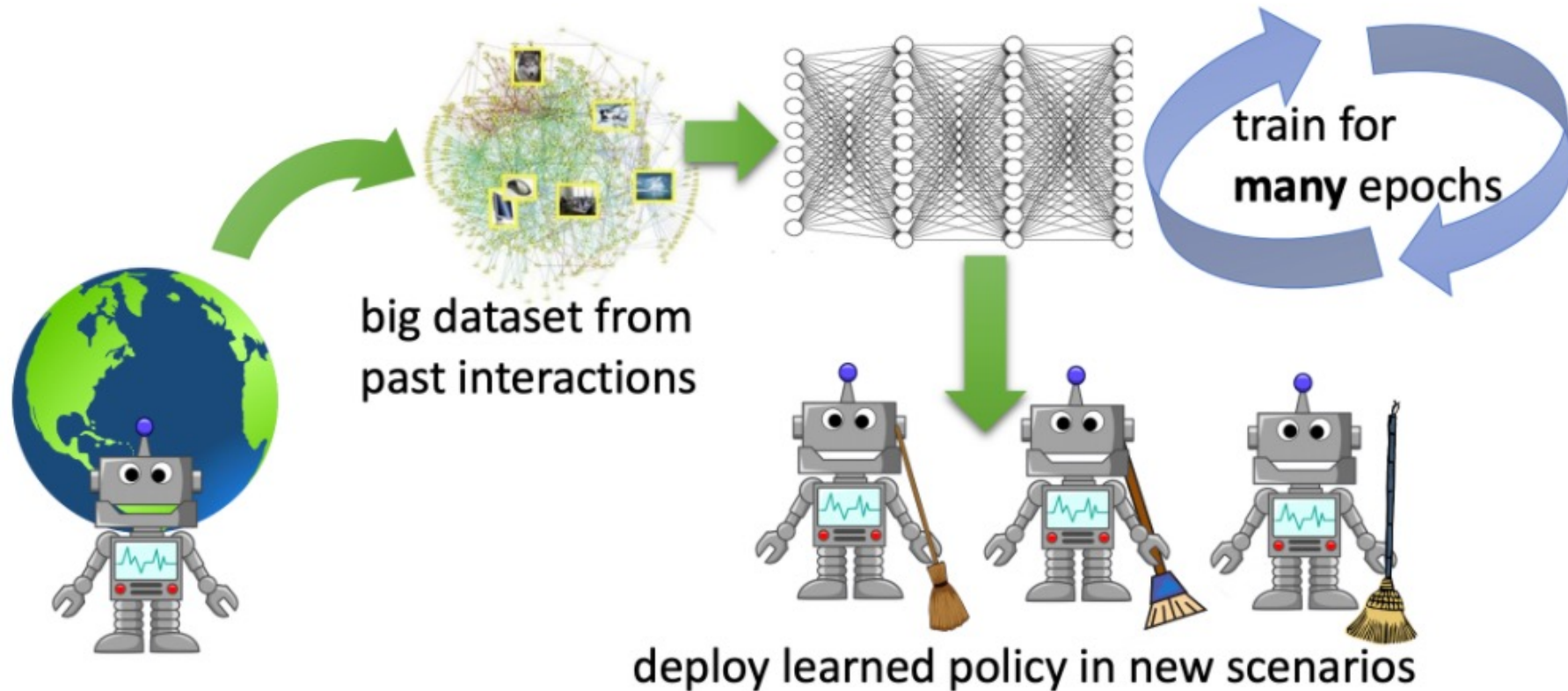
Drug discovery



Hits objects
unless properly
safeguarded

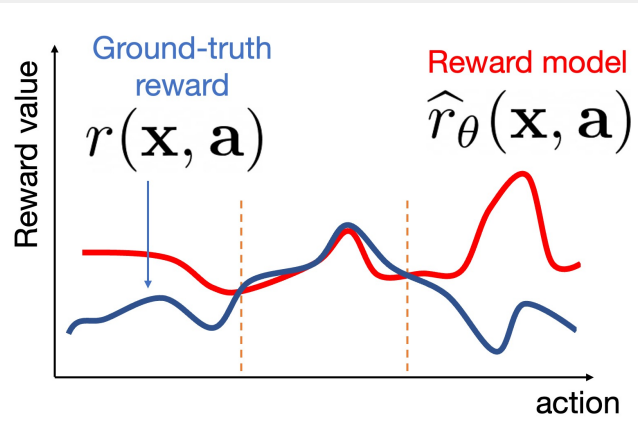
What Can We Do?

Solution: incorporate existing, static datasets



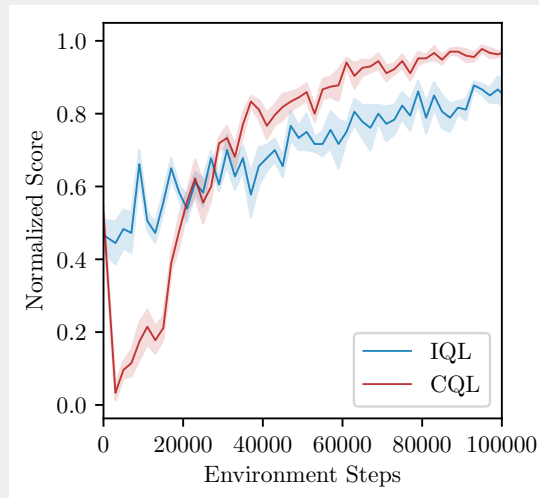
Today's Lecture: Several Facets of Offline RL

Algorithms for Offline RL



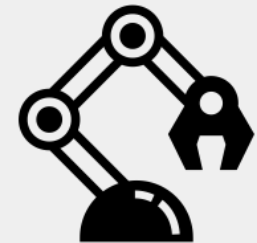
Conservative Q-learning,
Model-based offline RL,
large-scale results

Extensions of Offline RL



Online fine-tuning,
Offline RL with LLMs, etc

Applications in Robotics



Robotic pre-training + fine-tuning,
learning from videos

Notation & Terminology



Action from
the behavior policy

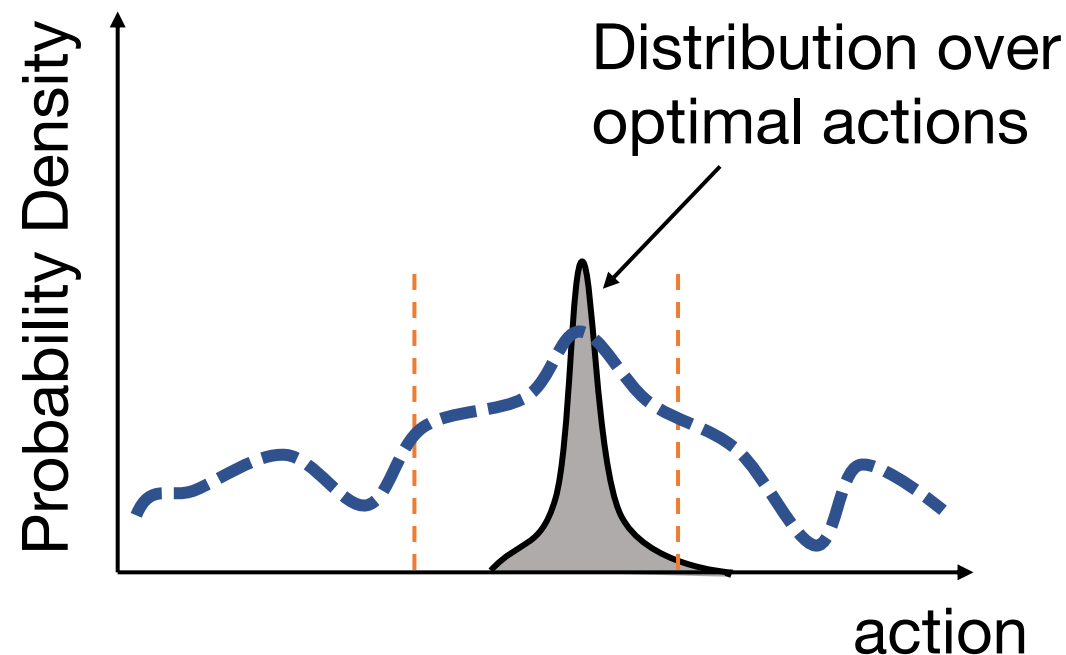
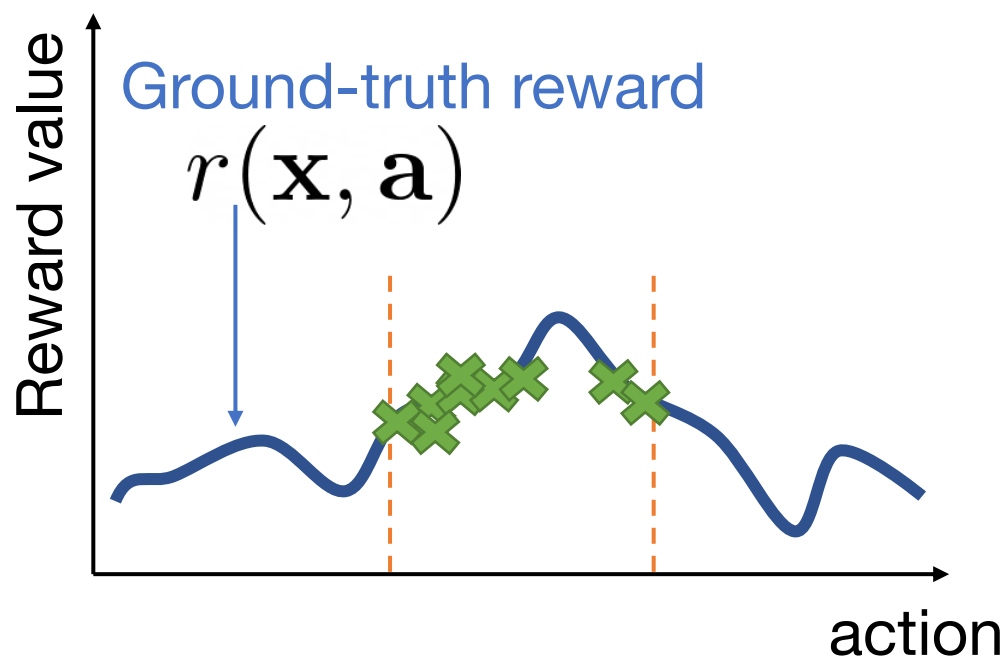
next
observation

$$\left\{ \underset{\text{observation}}{\mathbf{x}_i}, \underset{\text{Action from the behavior policy}}{\mathbf{a}_i}, \underset{\text{Reward value}}{r(\mathbf{x}_i, \mathbf{a}_i)}, \underset{\text{next observation}}{\mathbf{x}_{i+1}} \right\}_{i=1}^n$$

Visualization: One-Step Offline RL / Bandit

observation reward value

$$\mathcal{D} = \left\{ \left(\underset{\text{action}}{\mathbf{x}_i}, \underset{\text{action}}{\mathbf{a}_i}, \underset{\text{reward value}}{r(\mathbf{x}_i, \mathbf{a}_i)} \right) \right\}_{i=1}^n$$



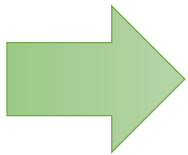
Starting Point: Off-Policy RL for Offline Bandits

observation

reward value

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{a}_i, r(\mathbf{x}_i, \mathbf{a}_i))\}_{i=1}^n$$

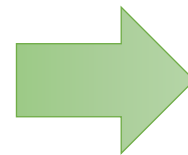
action



1

Fit a model to the
reward values

$$\hat{r}_\theta(\mathbf{x}, \mathbf{a})$$



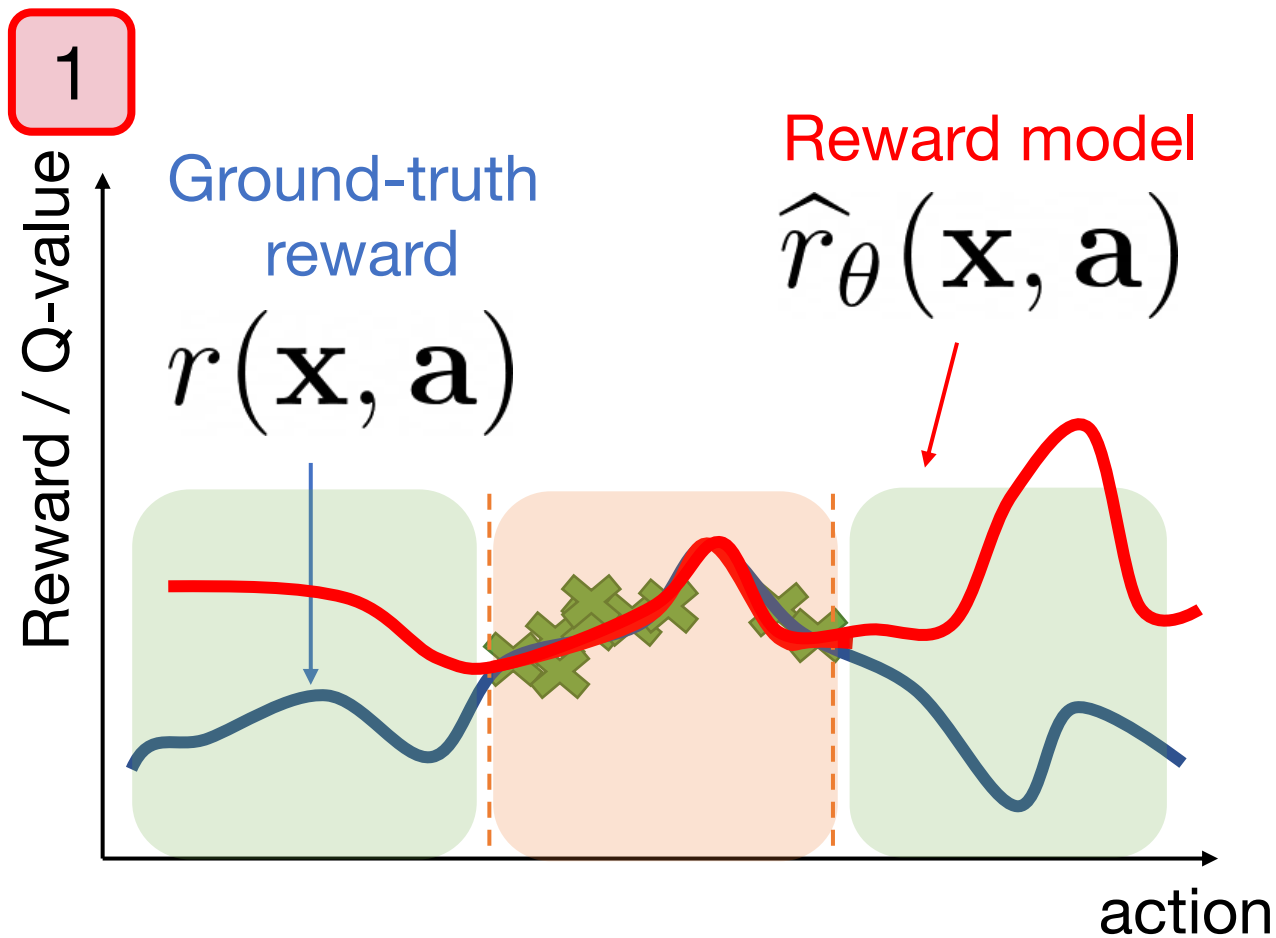
2

Train the policy to
maximize reward

$$\max_{\pi} \mathbb{E}_{\mathbf{a}_i \sim \pi(\cdot | \mathbf{x}_i)} [\cancel{r(\mathbf{x}_i, \mathbf{a}_i)}]$$

$$\hat{r}_\theta(\mathbf{x}_i, \mathbf{a}_i)$$

What Will the Reward Model Learn?



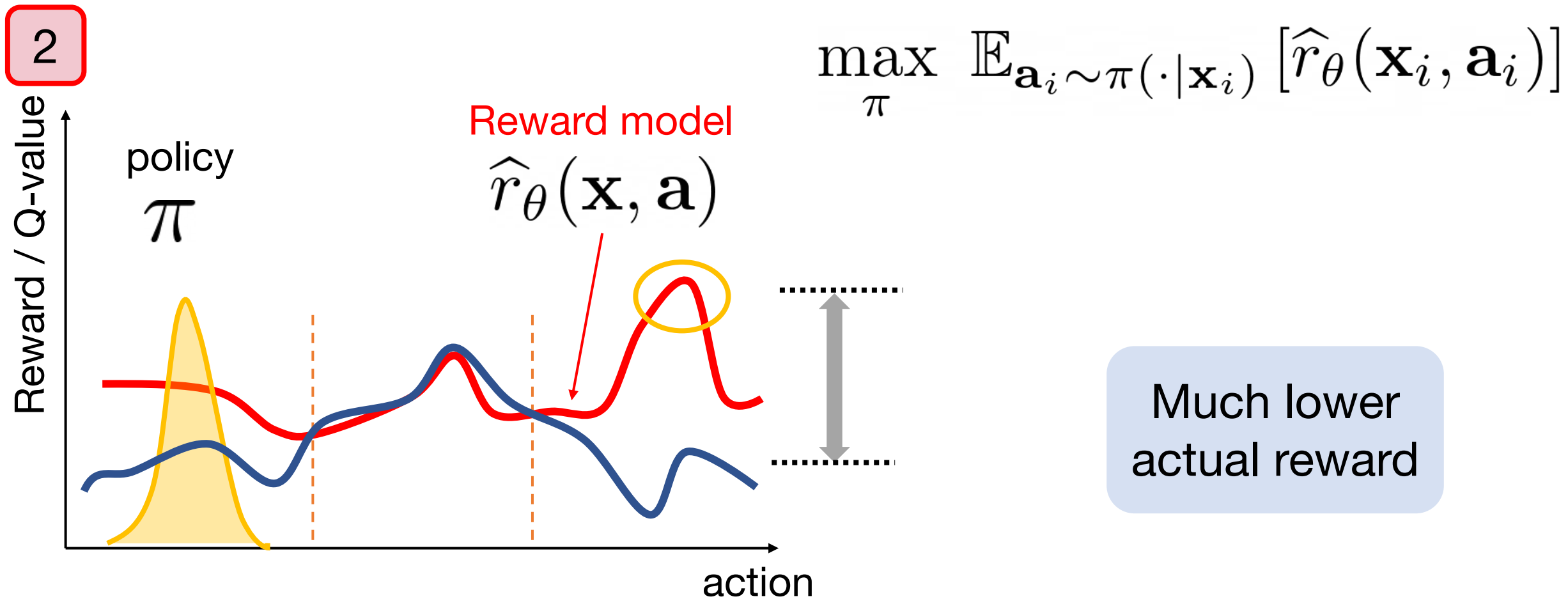
$$r(\mathbf{x}, \mathbf{a}) \approx \hat{r}_\theta(\mathbf{x}, \mathbf{a})$$

in expectation under $p_{\text{data}}(\mathbf{a}|\mathbf{x})$

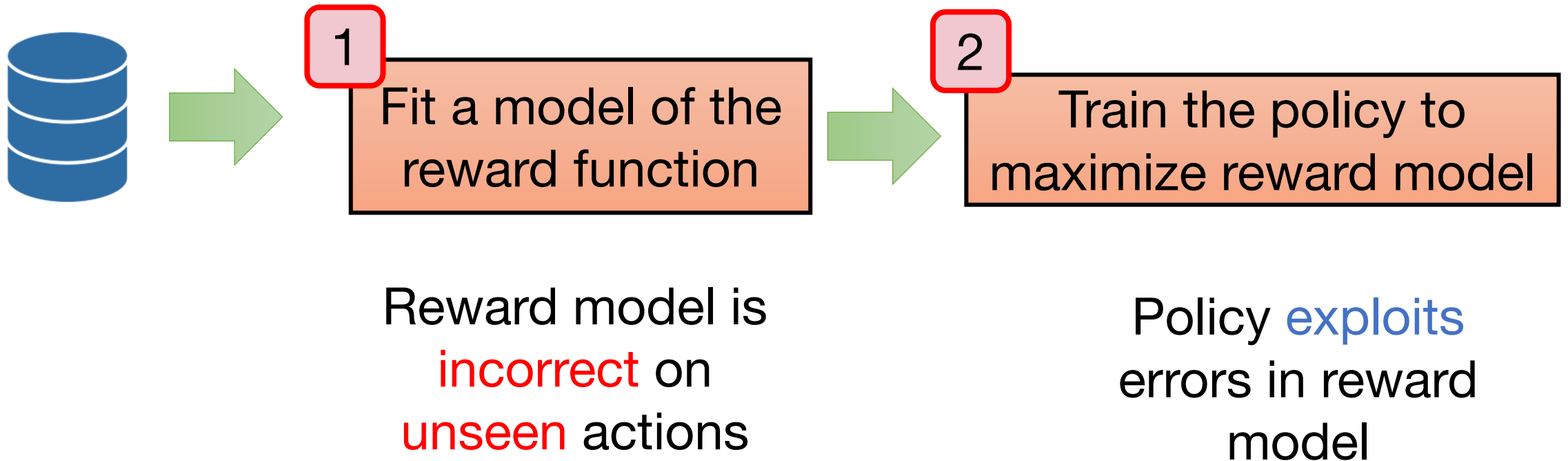
but not for
unseen actions

Distribution shift

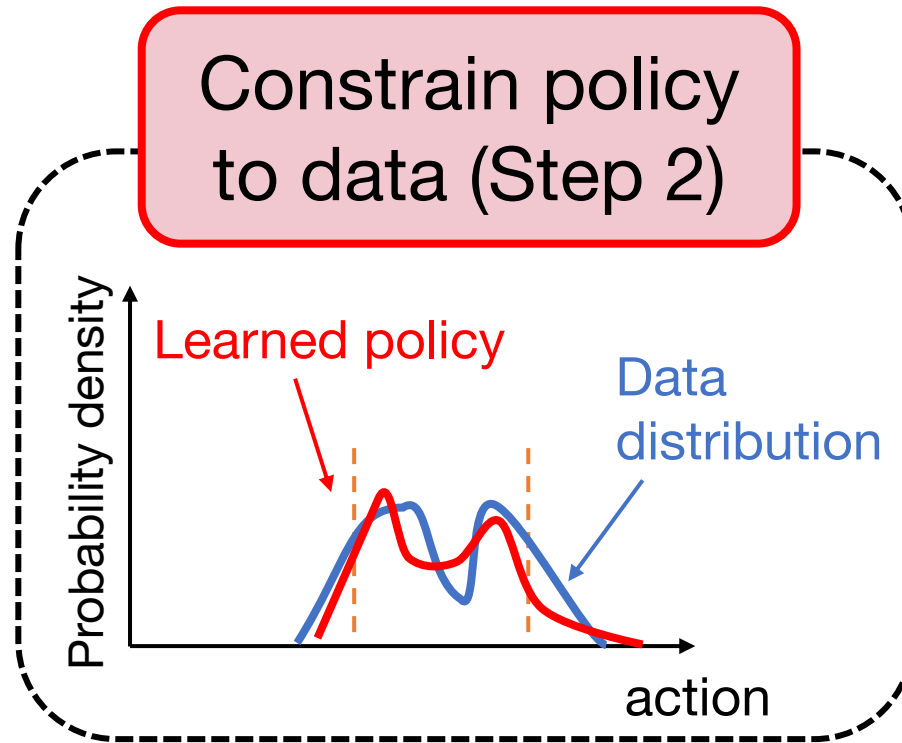
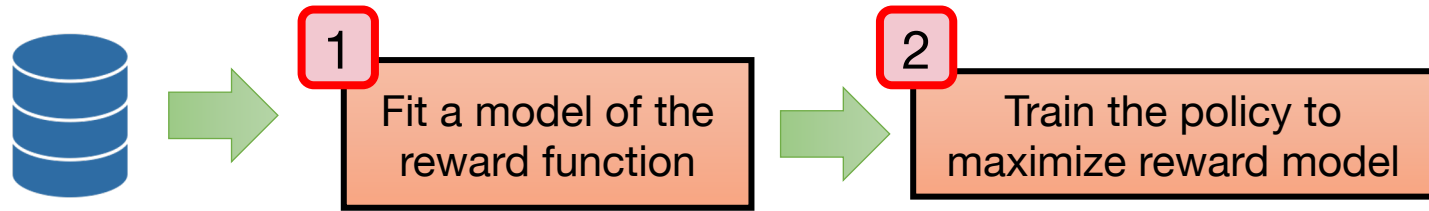
Which Policy Will Optimization Find?



Summary: The Key Challenge in Offline RL



Prior Approaches



Can we address
Step 1 directly?

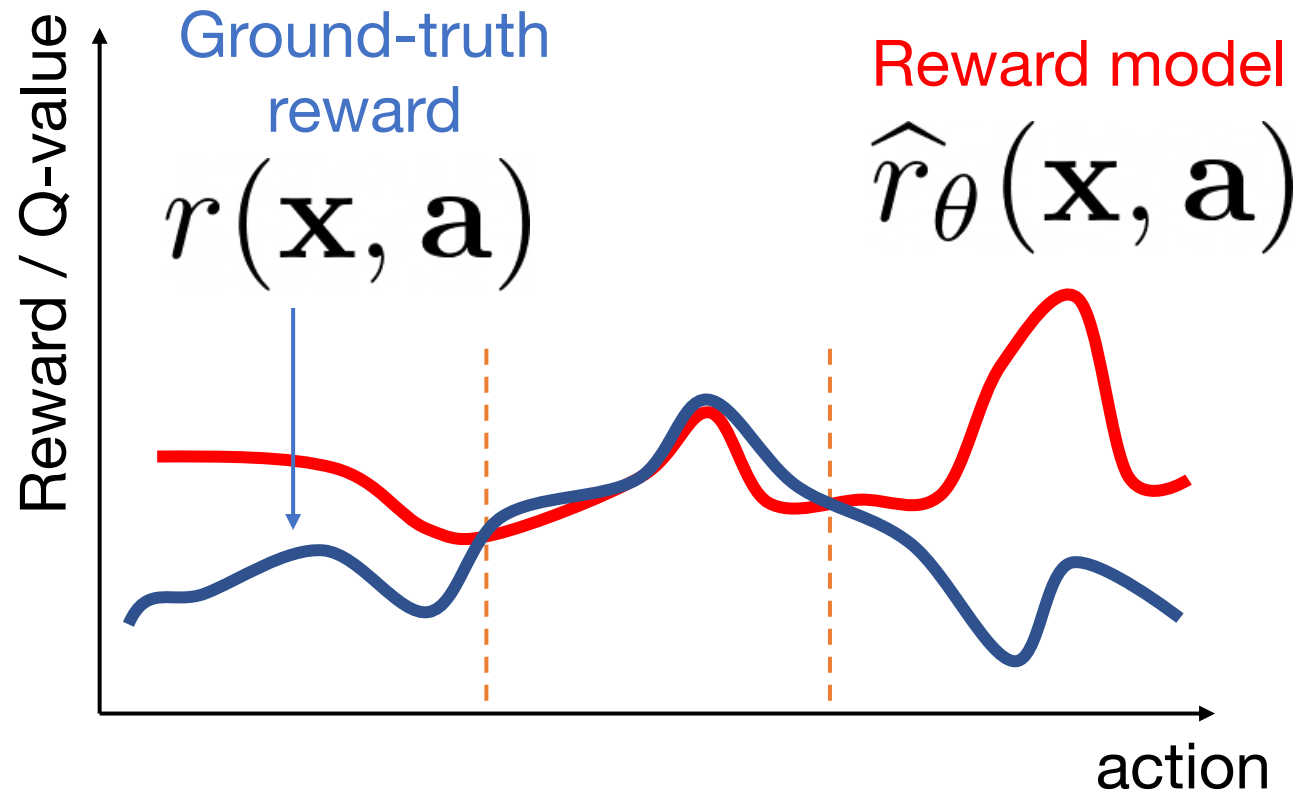
Policy constraints

(Fujimoto et al. 2018; [Kumar et al. 2019](#); [Peng et al. 2019](#); Wu et al. 2019, Jaques et al. 2019; Laroché et al. 2020; Liu et al. 2020; Kostrikov et al. 2020, 2021)

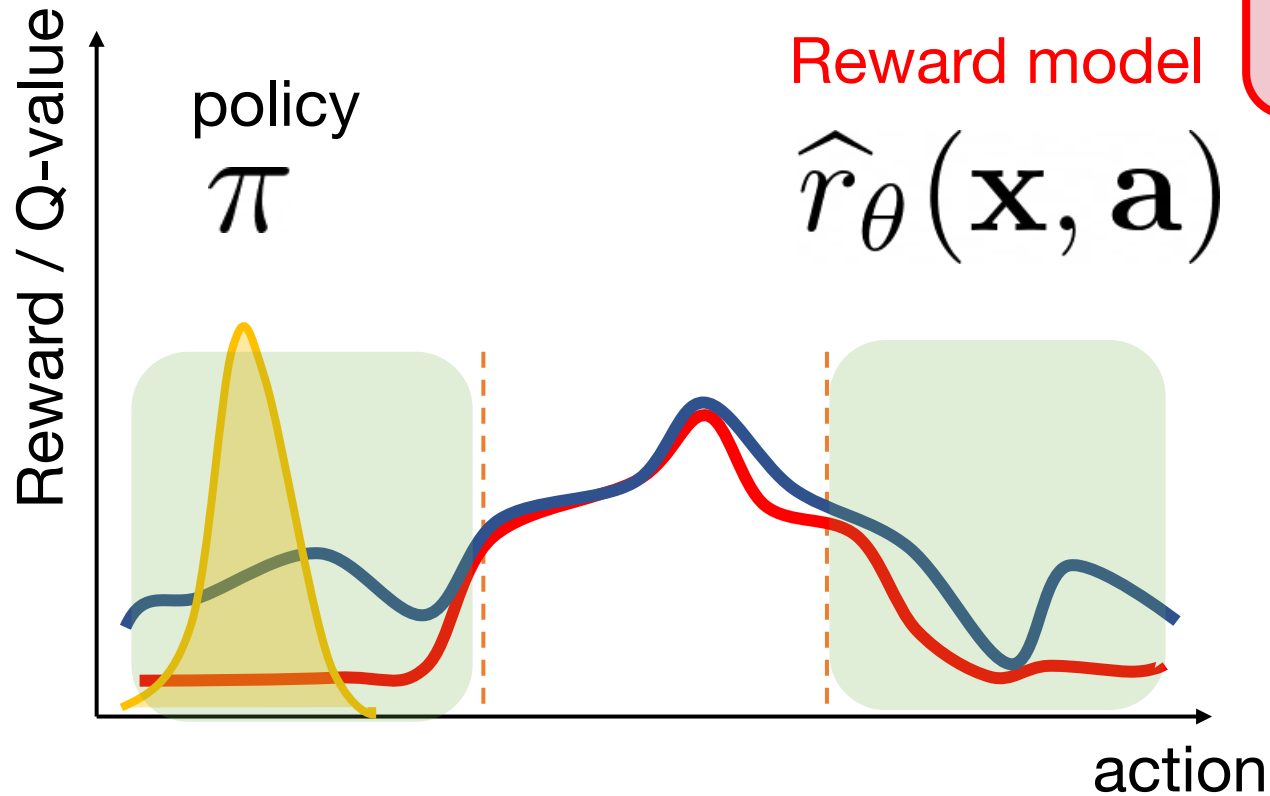
State-marginal constraints

(Uehara & Jiang, 2019; Nachum et al. 2019; Zhang et al. 2020; Nachum et al. 2020)

Idea: Which Reward Models Do Not Hurt?



Idea: Which Reward Models Do Not Hurt?



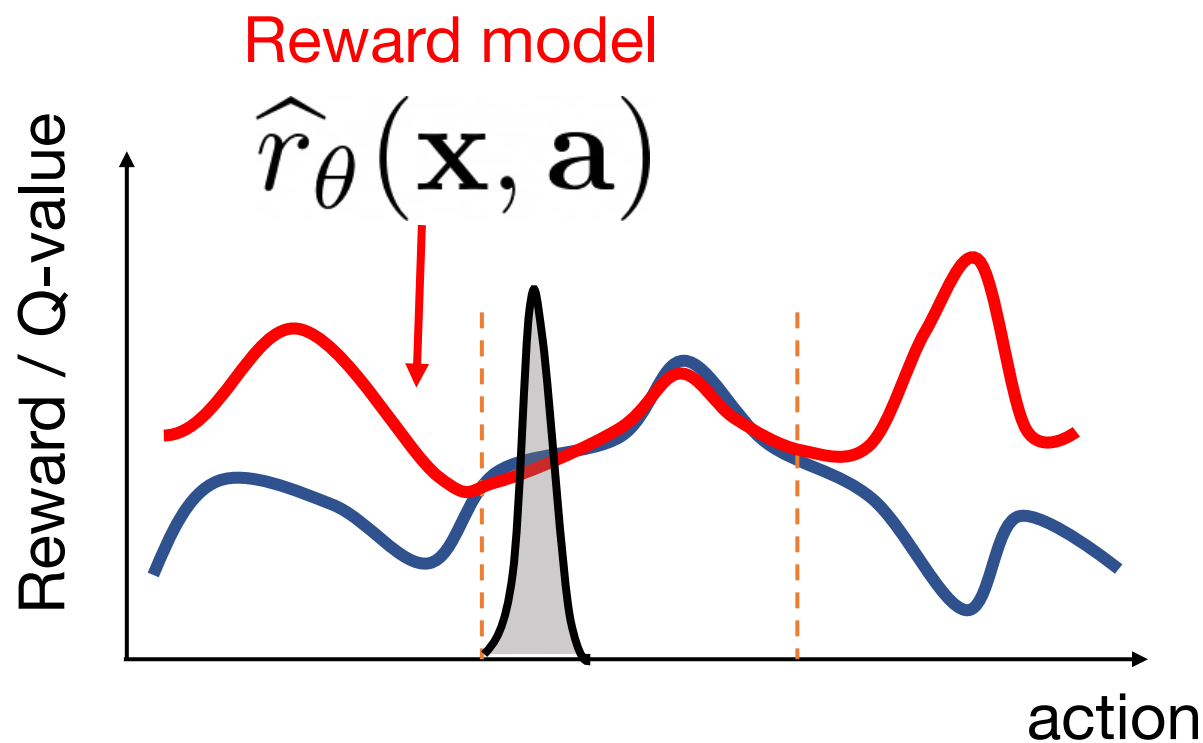
Key Contribution: A method to learn such a conservative model

if we can ensure
$$\hat{r}_\theta(\mathbf{x}, \bar{\mathbf{a}}) \leq r(\mathbf{x}, \bar{\mathbf{a}})$$

on unseen actions

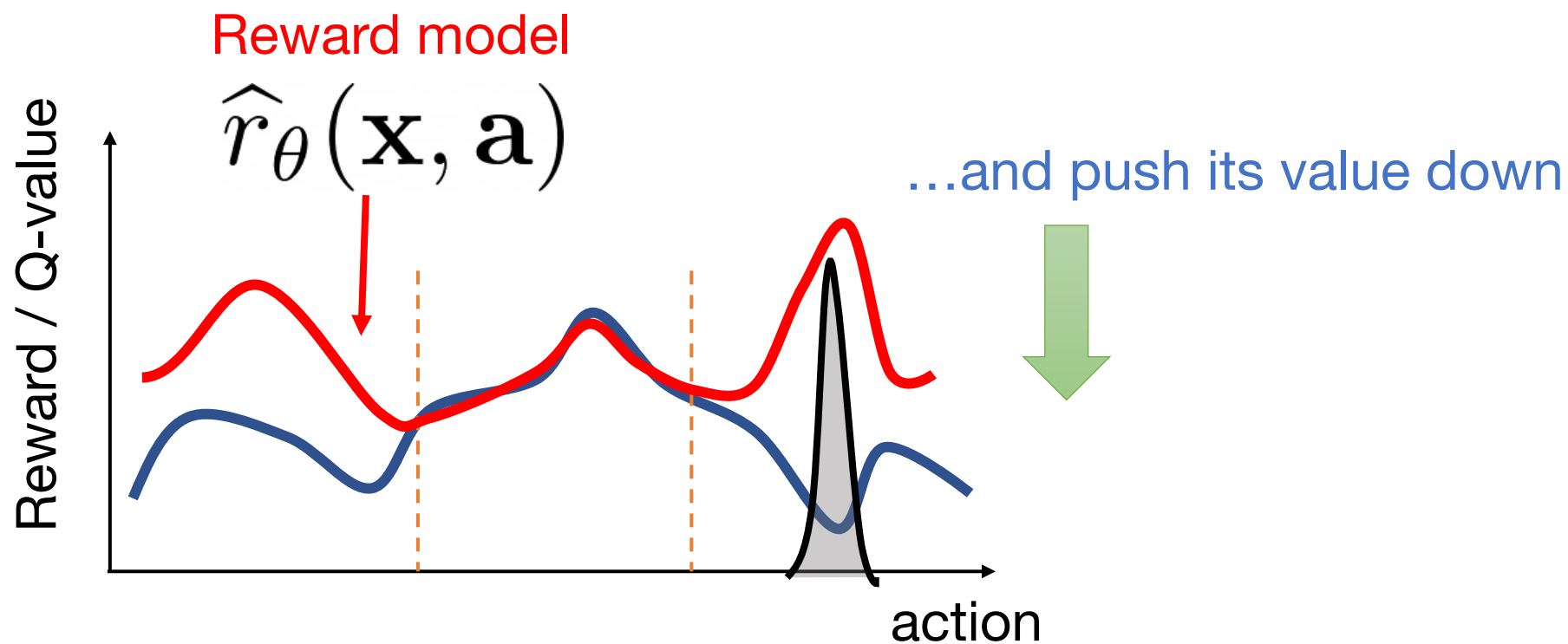
Learning Conservative Models for Offline RL

Find an **unseen** action that **maximizes** the learned reward model



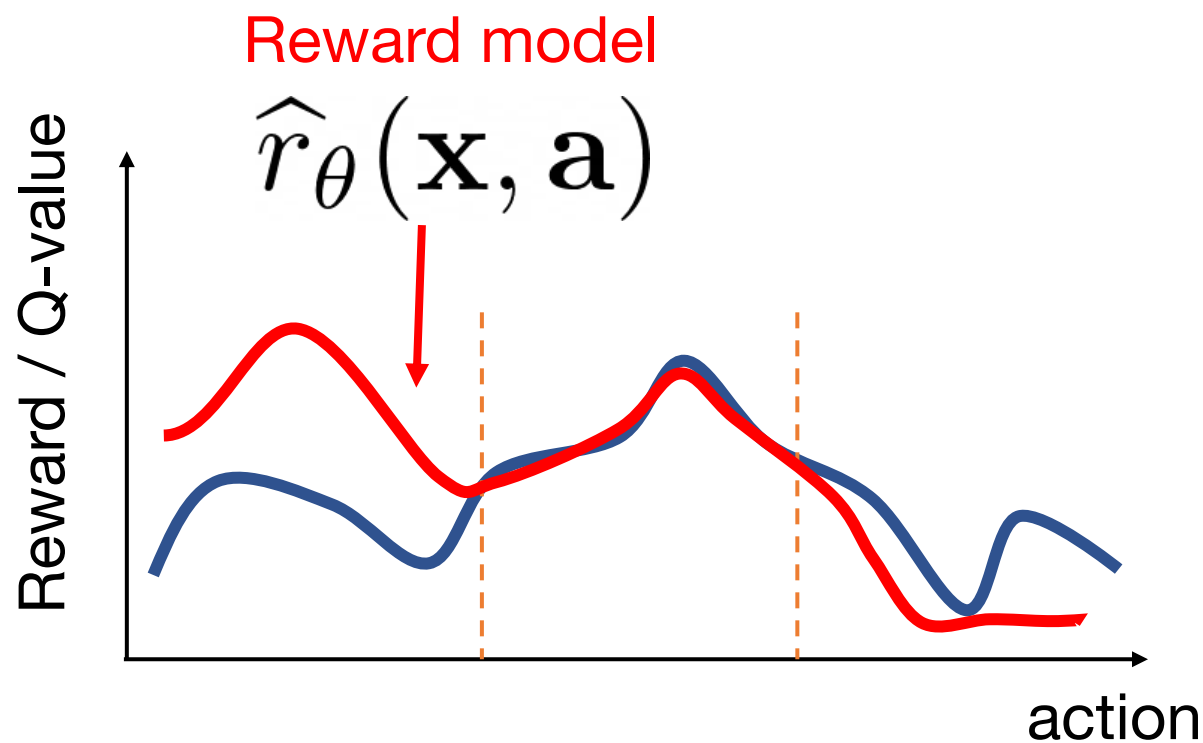
Learning Conservative Models for Offline RL

Find an **unseen** action that **maximizes** the learned reward model



Learning Conservative Models for Offline RL

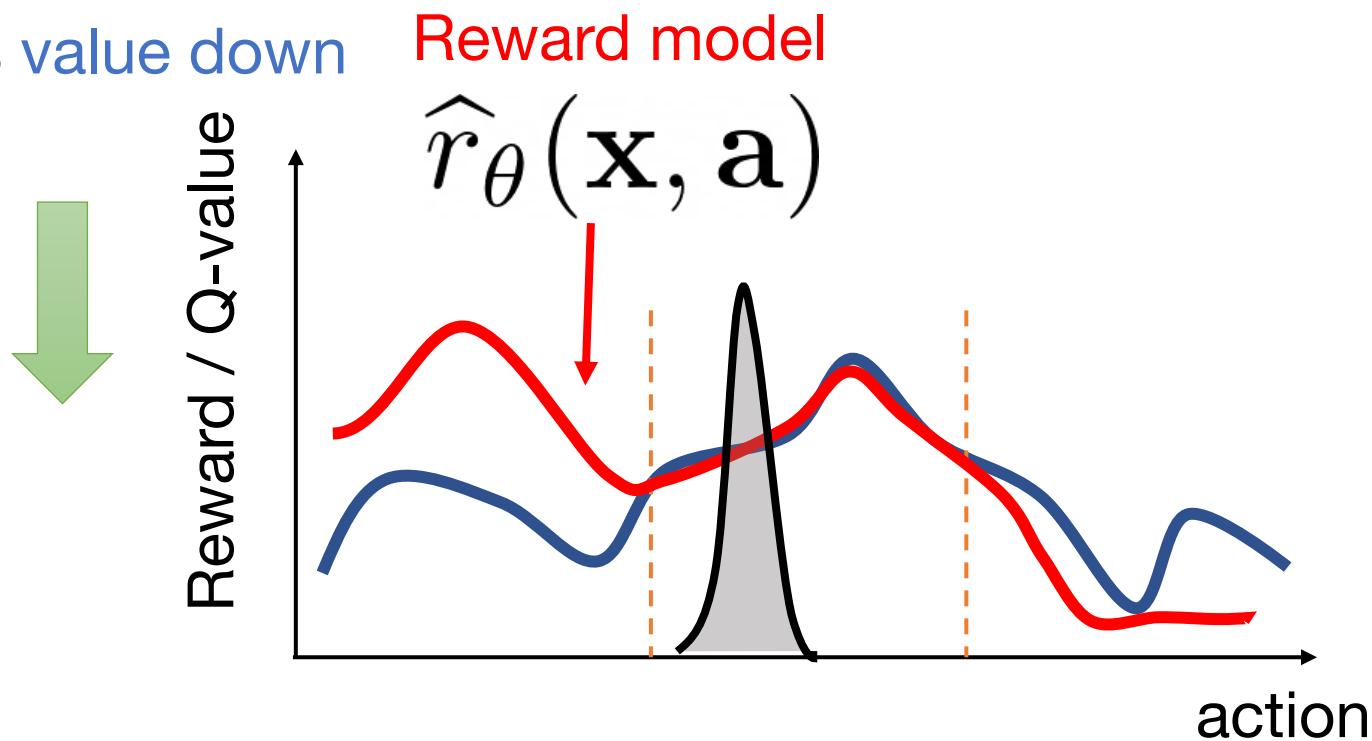
Find an **unseen** action that **maximizes** the learned reward model



Learning Conservative Models for Offline RL

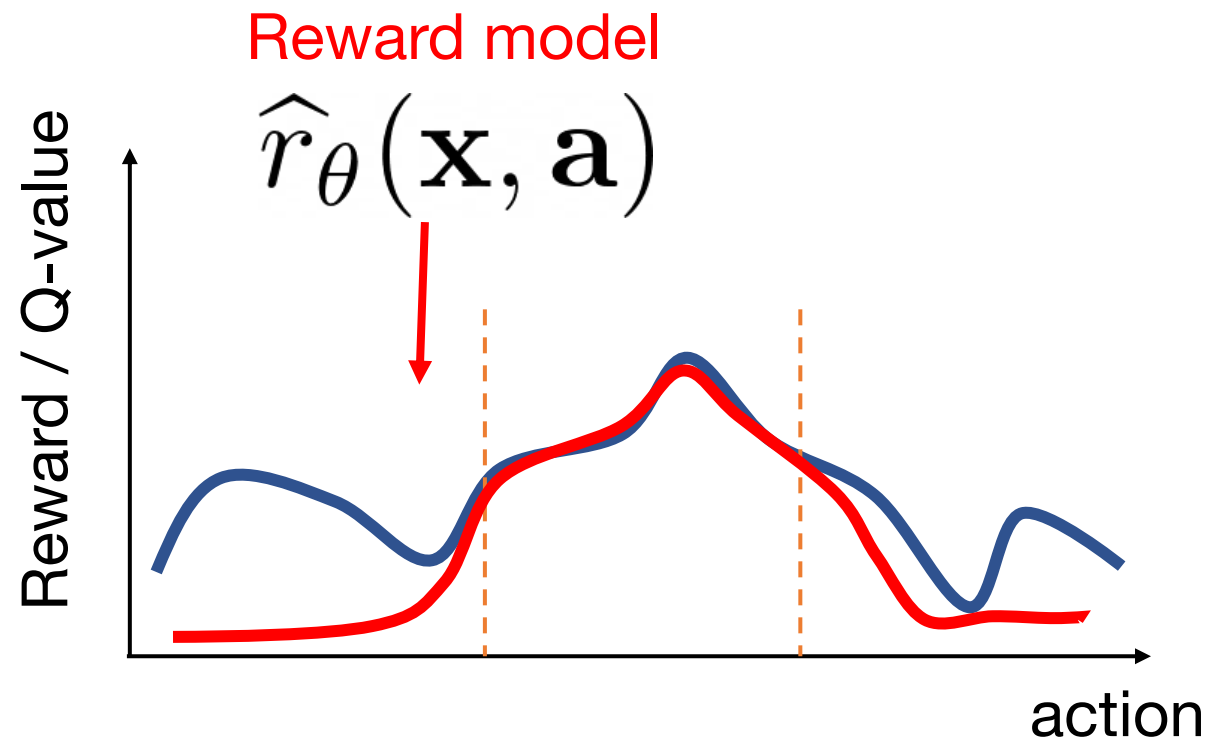
Find an **unseen** action that **maximizes** the learned reward model

...and push its value down



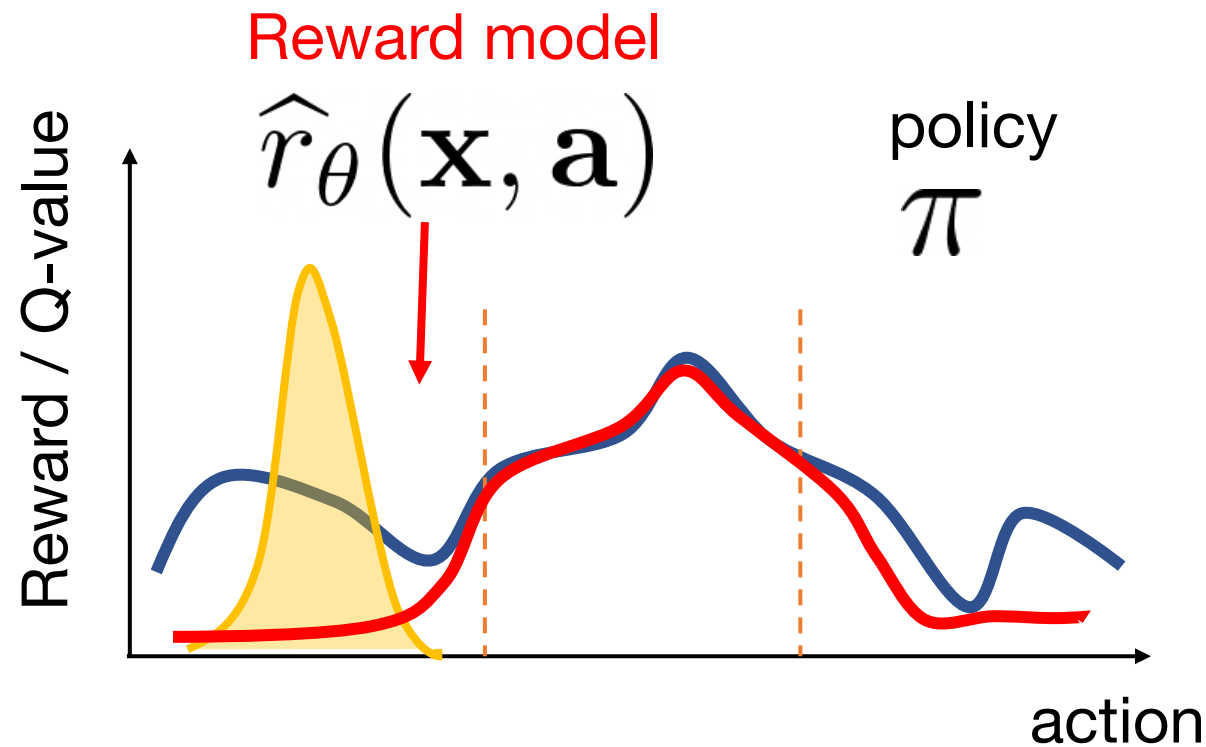
Learning Conservative Models for Offline RL

Find an **unseen** action that **maximizes** the learned reward model

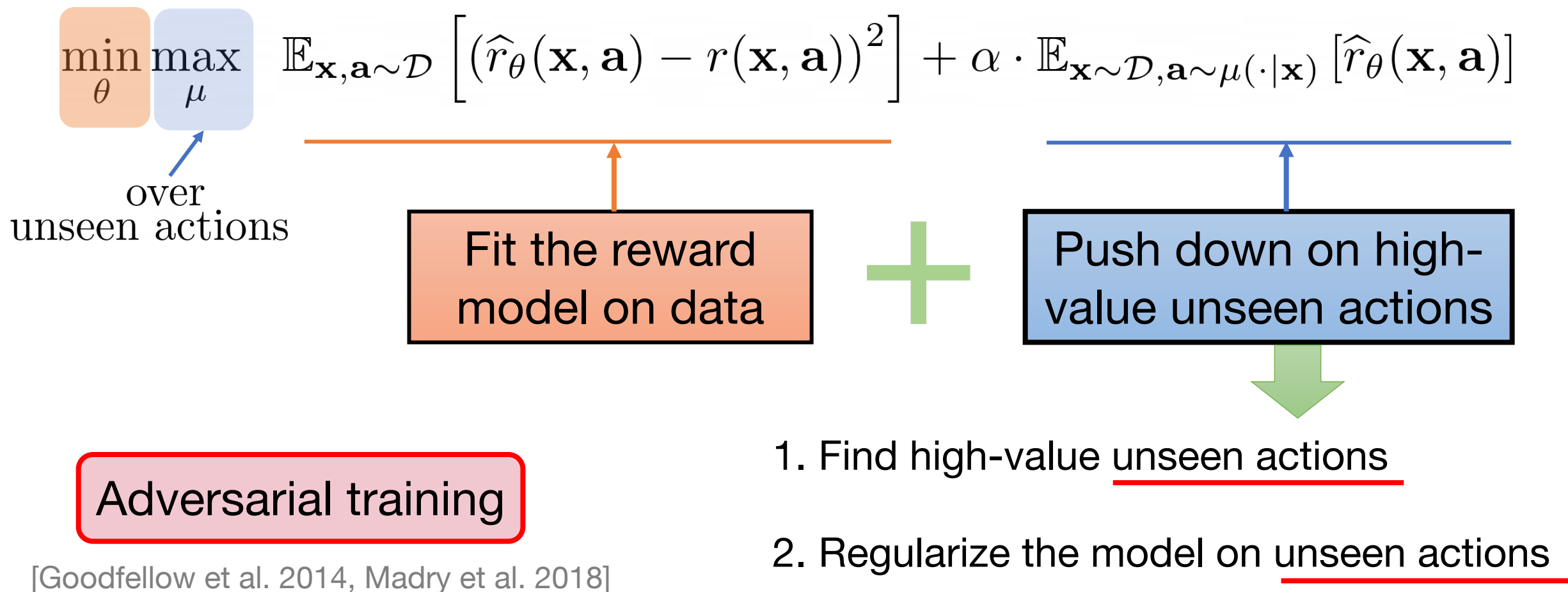


Policy Optimization (Step 2) Now Works

Conservative model



Summary: Conservative Offline Bandits



Conservative Q-Learning

Reward model

~~$\hat{r}_\theta(\mathbf{x}, \mathbf{a})$~~

Cumulative rewards
under the policy

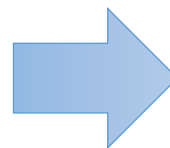
$$\sum_{i=1}^T r(\mathbf{x}_i, \mathbf{a}_i)$$

More severe exploitation
because target values
depend on policy



Q-function

$$\hat{Q}_\theta(\mathbf{x}, \mathbf{a})$$



Conservative Q-Learning (CQL)

=

Fit Q-function
on data



Push down high-value
unseen actions

Practical Version: Conservative Q-Learning

$$\min_{\theta} \max_{\mu} \alpha \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{a}' \sim \mu} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}')] + \mathbb{E}_{\mathbf{x}, \mathbf{a}, \mathbf{x}' \sim \mathcal{D}} \left[\left(\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}) - (r(\mathbf{x}, \mathbf{a}) + \gamma \hat{Q}_{\theta}(\mathbf{x}', \mathbf{a}')) \right)^2 \right]$$

over
unseen actions

$$\min_{\theta} \max_{\mu} \alpha \left(\mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{a}' \sim \mu} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}')] - \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a})] \right) + \text{TD-Error}(\theta)$$

relative gaps, but no restriction on adversary

$$\min_{\theta} \max_{\mu} \alpha \left(\mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{a}' \sim \mu} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}')] - \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a})] \right) + \text{TD-Error}(\theta) + \mathcal{H}(\mu)$$

smooth out the adversary

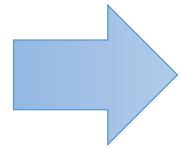
$$\min_{\theta} \alpha \left(\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\log \sum_{\mathbf{a}' \in \mathcal{A}} \exp \left(\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}') \right) \right] - \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a})] \right) + \text{TD-Error}(\theta)$$

Behavioral cloning loss

Extension to Model-Based RL

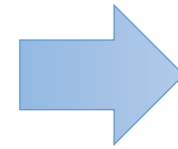
Cumulative rewards
under the policy

$$\sum_{i=1}^T r(\mathbf{x}_i, \mathbf{a}_i)$$



Learned
world model

$$\hat{p}_{\phi}(\mathbf{x}' | \mathbf{x}, \mathbf{a})$$



Q-function

$$\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a})$$

**Conservative
model-based RL**

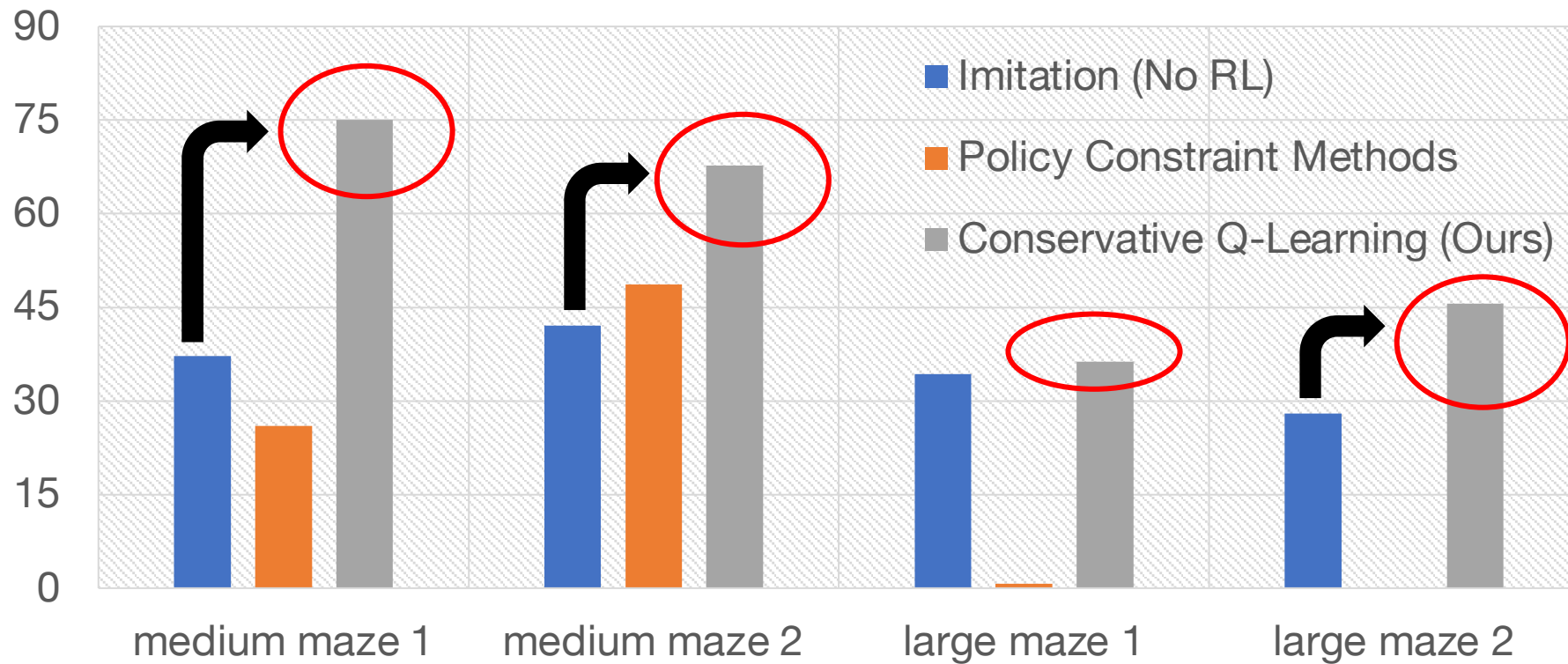
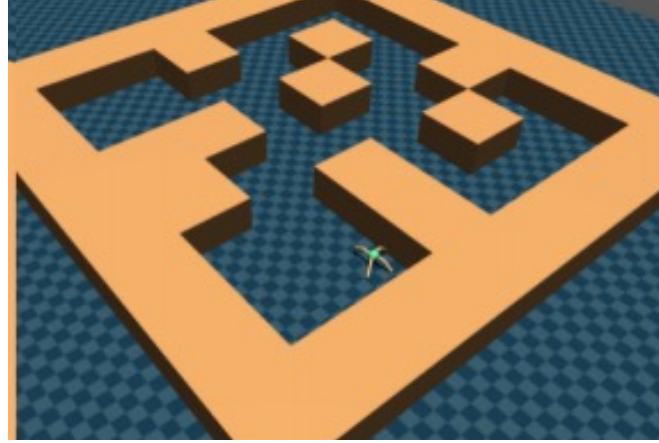
=

Fit Q-function
on model data

+

Push down high-
value unseen states

Benchmark Results



Adversarially-Trained
Actor-Critic for Offline RL

Cheng*, Xie*,
Jiang, Agarwal. ICML 2022

COMBO: Conservative
Offline Model-Based RL

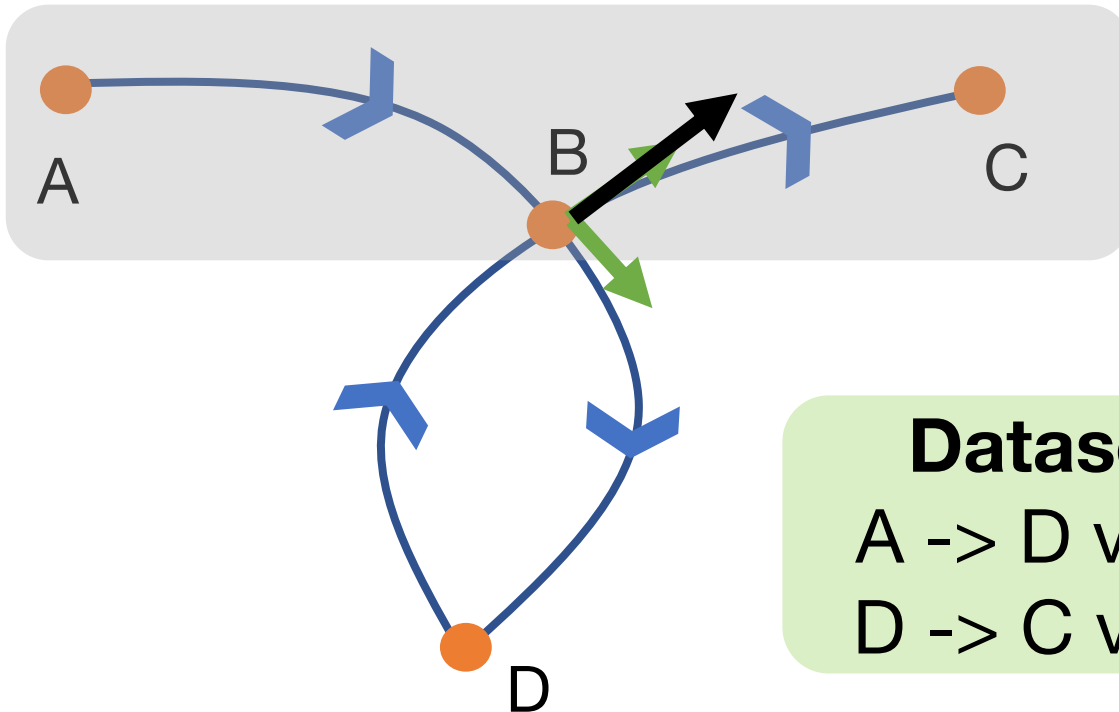
Yu*, **Kumar*** et al.
NeurIPS 2021

Intuition: Why *Can* Offline RL Improve?

Key idea: Compose the **best actions** from sub-optimal sequences to improve

Imitation:
50% of the time
wrong action at B

Offline RL:
Finds path better than
data, **by stitching**



Dataset:
A -> D via B
D -> C via B

***Scaling Models
in Offline RL***

Large Models in Offline Deep RL

Stop Regressing: Training Value Functions via Classification for Scalable Deep RL. arXiv 2024.
Offline RL on Diverse Multi-Task Data Both Scales and Generalizes. ICLR 2023 (Oral)

DR3: Value-Based RL Requires Explicit Regularization. ICLR 2022 (Spotlight)

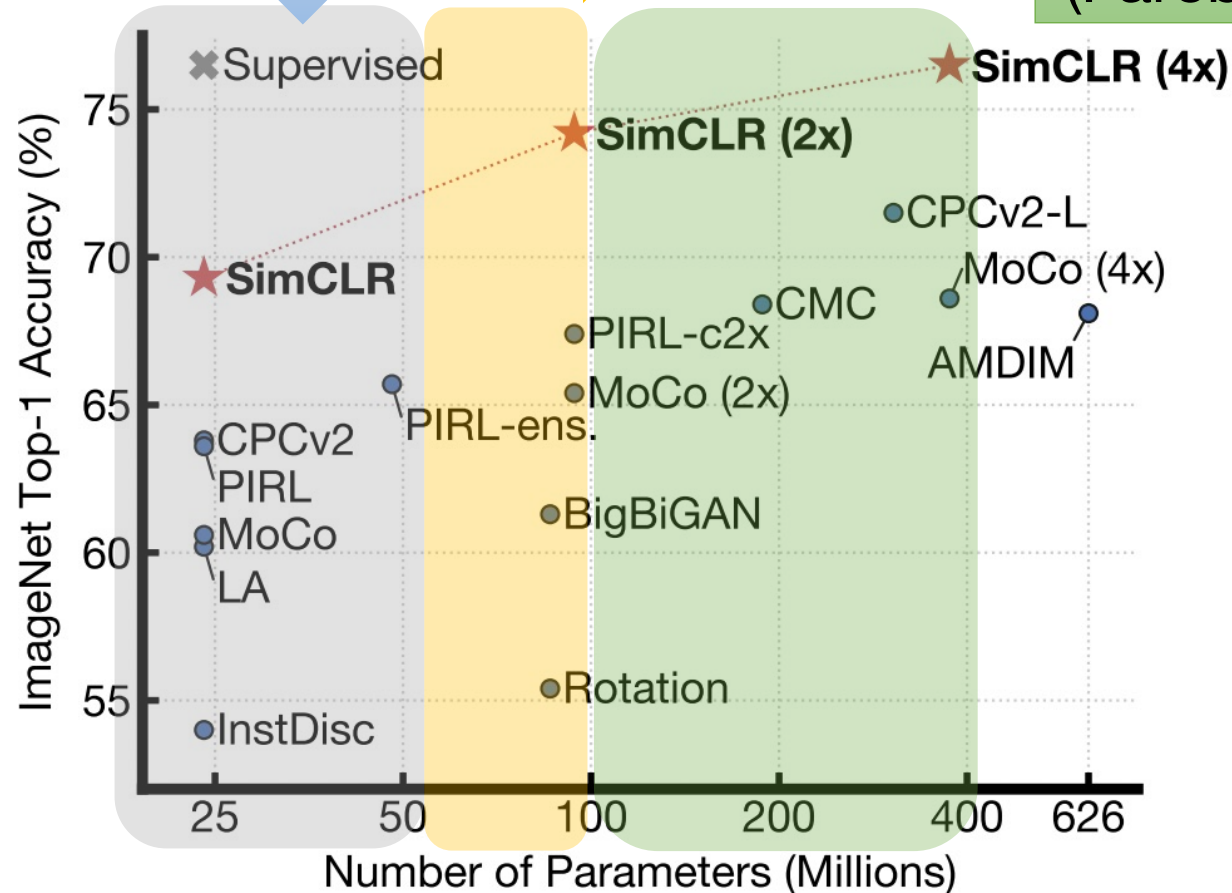
Status of Scaling in RL

Our initial work
(Kumar et al. ICLR 2023)

Our follow-up work
(Farebrother et al. 2024)

Prior work in RL:

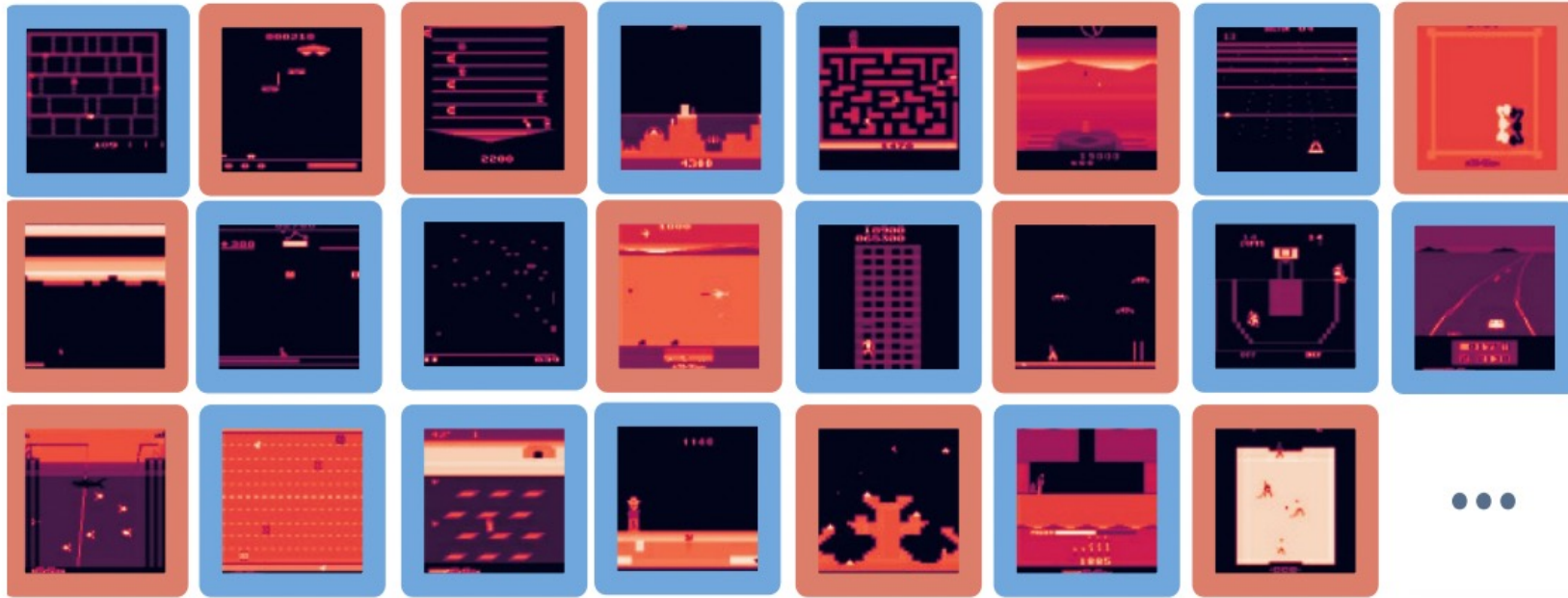
AlphaGo Zero
AlphaStar
MuZero ReAnalyze



Picture from:

Chen, Kornblith, Norouzi, Hinton. **A Simple Framework for Contrastive Learning of Visual Representations.** ICML 2020.

Large-Scale Training of Generalist Policies



Train a **single policy** on 40 Atari games

2 billion data-points,
40 games,
sub-optimal data

Lee et al. **Multi-Game Decision Transformers**. NeurIPS 2022 (Oral)

Also related: **Agent 57: Outperforming the Human Atari Benchmark**. DeepMind, 2020.

Kumar et al. **Offline RL on Diverse Multi-Task Data Both Scales and Generalizes**. ICLR 2023 (Notable top 5%)

2 billion data-points,
40 games,
sub-optimal data



Ingredients For Scaling Offline Q-Learning

An offline RL algorithm

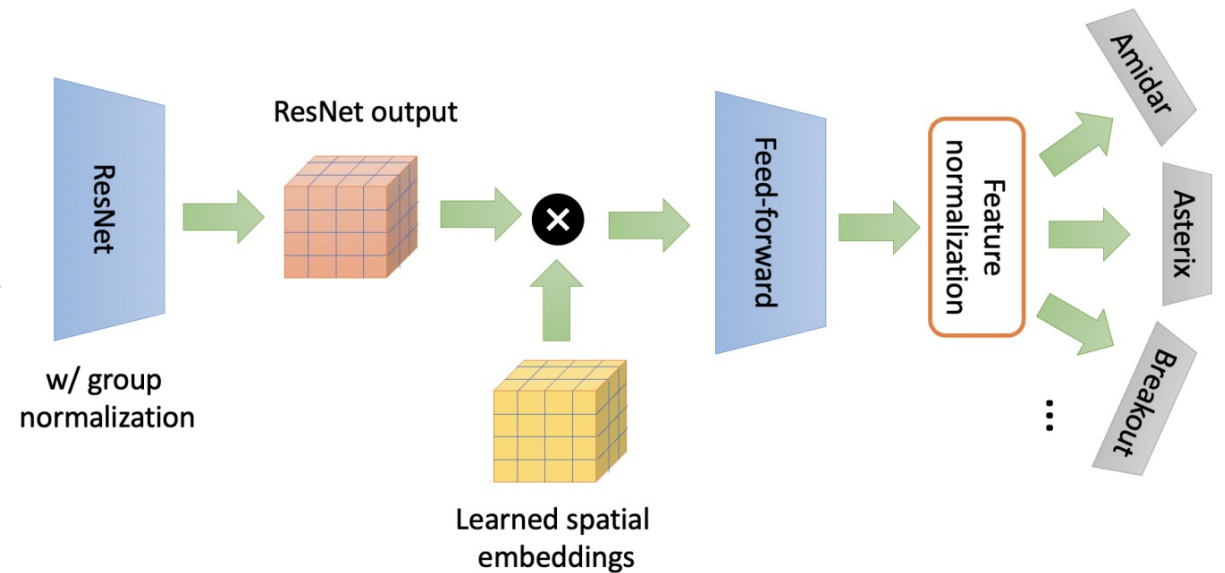
Conservative Q-learning (CQL)

A large neural network

A way to effectively use capacity

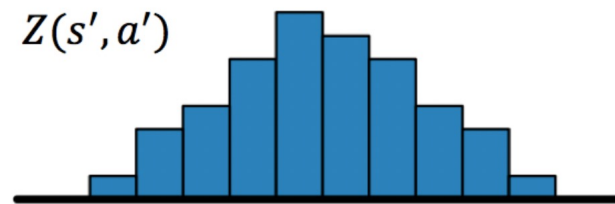
DR3 normalization: Normalize features to have norm = 1

$$\text{Normalize}(\phi(s, a)) = \frac{\phi(s, a)}{\|\phi(s, a)\|_2}$$



Q-value discretization (distributional RL)

$Z(s', a')$



Q-value = categorical distribution

Cross-entropy loss

Why is it hard to train large models with offline RL?

Why is it easy to train large models with supervised learning?

Implicit regularization

$\min_{\theta} \mathcal{L}(\theta)$

Over-parameterized $\nearrow$

Stochastic gradient descent $\downarrow$

$\min_{\theta} \mathcal{L}(\theta) + \mathcal{R}(\theta)$

$\nwarrow$

“implicit regularizer”

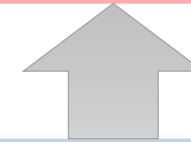
(prefers simpler solutions)

Why is it hard to train large models with offline RL?



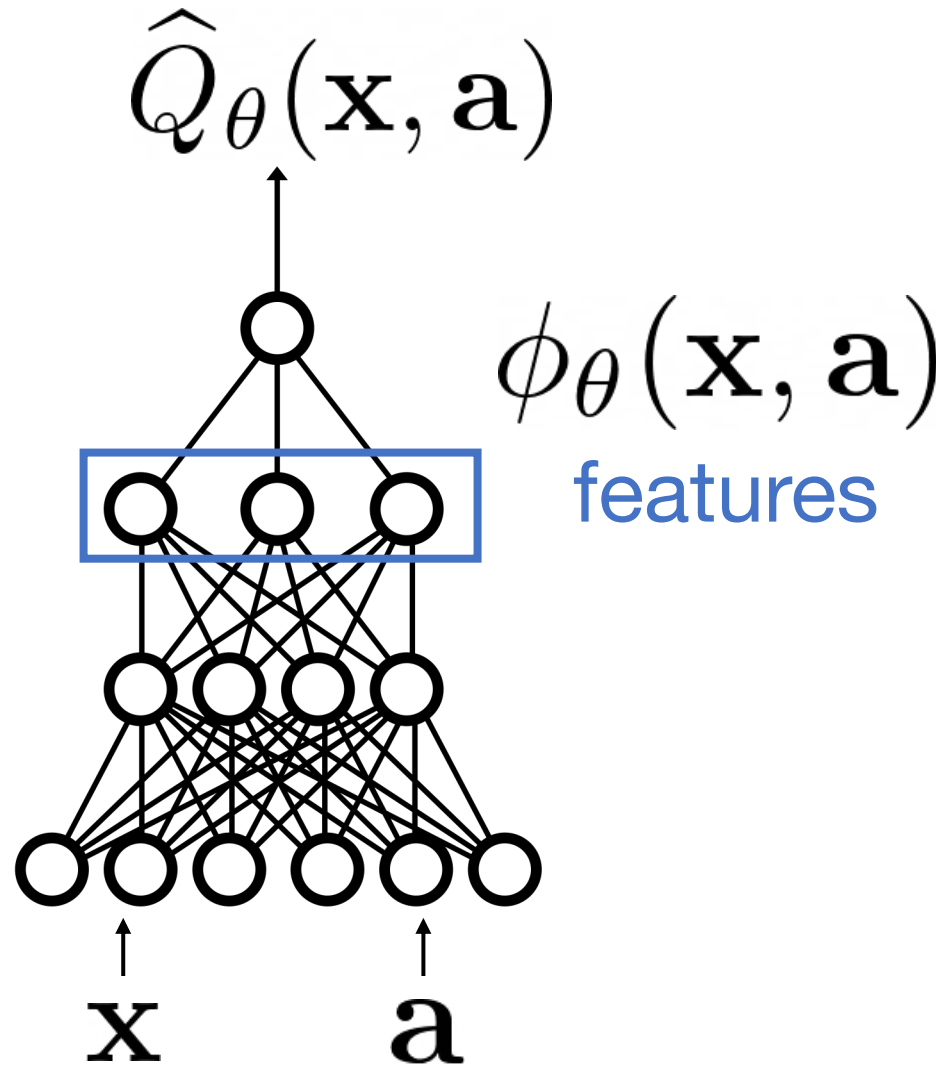
Why is it easy to train large models with supervised learning?

Implicit regularizer in offline RL??



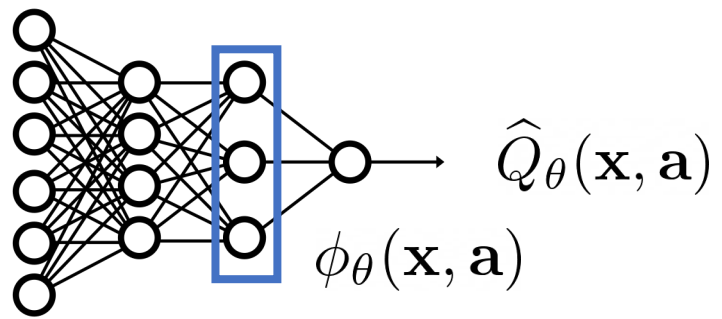
Implicit regularizer
prefers “simple” solutions
that perform well

Setup: Implicit Regularization in Offline RL



Theory: Implicit Regularization in Offline RL

[Informal] Theorem (Implicit regularization in SL vs offline RL)



Theory: Implicit Regularization in Offline RL

[Informal] Theorem (Implicit regularization in SL vs offline RL)

$$\mathcal{R}_{\text{SL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2$$

Supervised learning (SL)

[Blanc et al. COLT 2021]

$$\mathcal{R}_{\text{RL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2 -$$

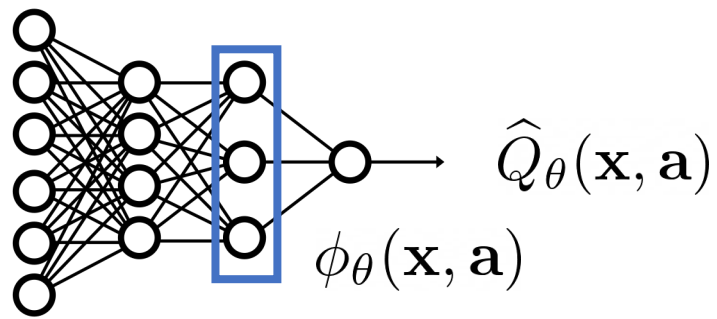
$$\sum_{i=1}^n \mathbb{E}_{\mathbf{a}'_{i+1} \sim \pi} \left[\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)^{\top} \phi_{\theta}(\mathbf{x}_{i+1}, \mathbf{a}'_{i+1}) \right]$$

Offline RL

[Kumar et al. ICLR 2022]

Reduce feature
lengths / norms

Conflicts by blowing
up feature norms

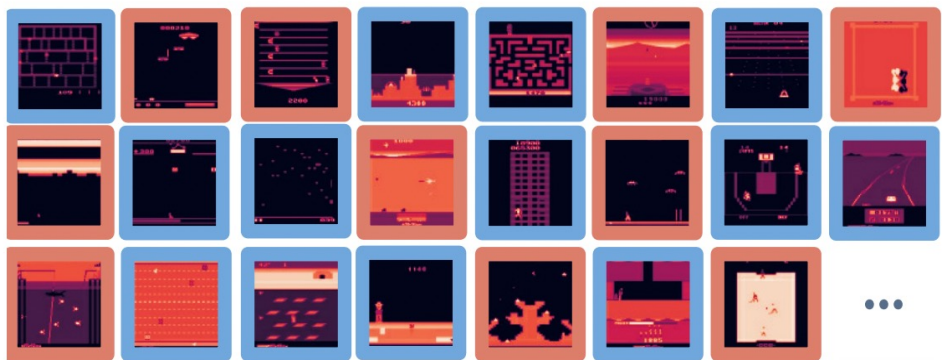


Implicit Reg. in RL → Large Feature Dot Products

$$\mathcal{R}_{\text{SL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2$$
$$\mathcal{R}_{\text{RL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2 - \sum_{i=1}^n \mathbb{E}_{\mathbf{a}'_{i+1} \sim \pi} [\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)^{\top} \phi_{\theta}(\mathbf{x}_{i+1}, \mathbf{a}'_{i+1})]$$

Key Insight: As this term is bad, control this explicitly during RL

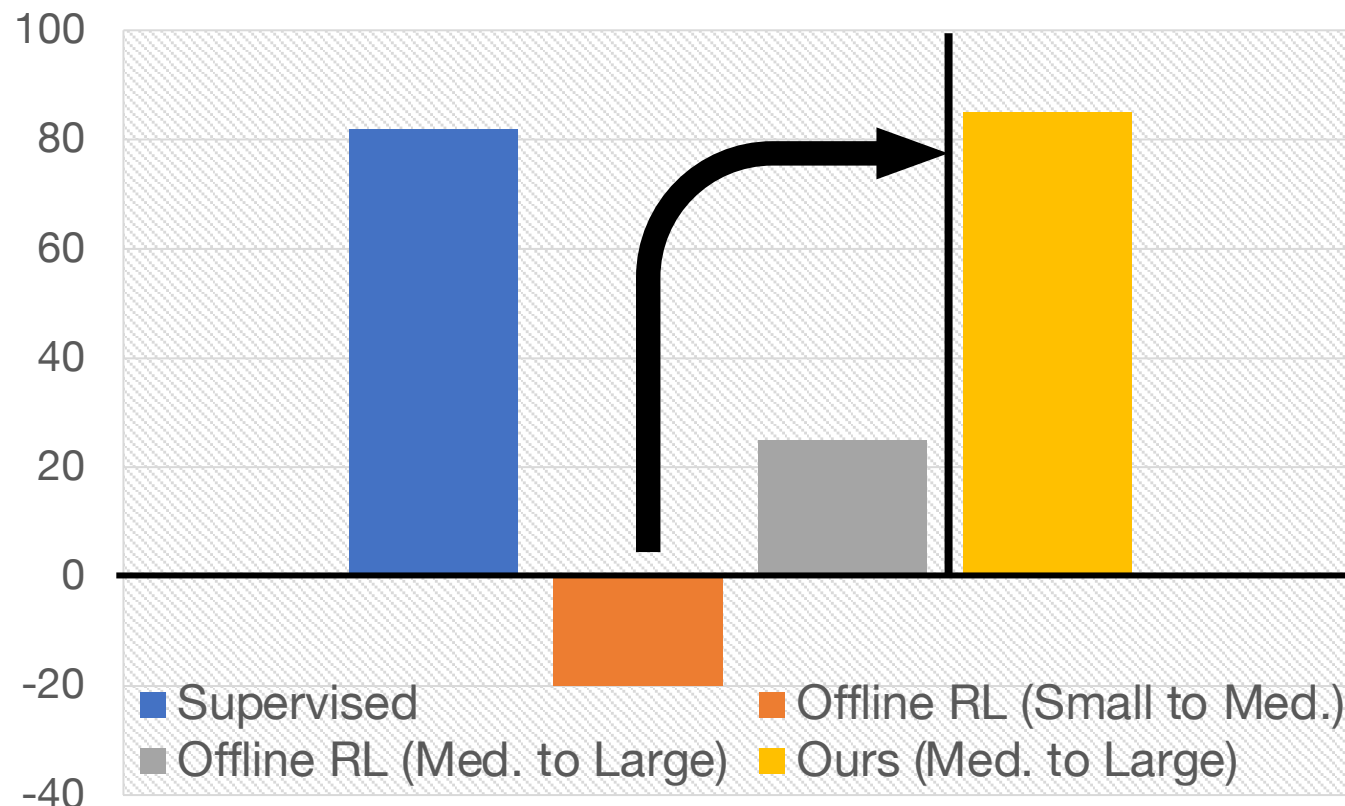
Regularizer Improves Scalability of Offline RL



Train a **single policy** on 40 Atari games

2 billion data-points,
40 games,
sub-optimal data

**% performance improvement with
model size (Small → Large)**



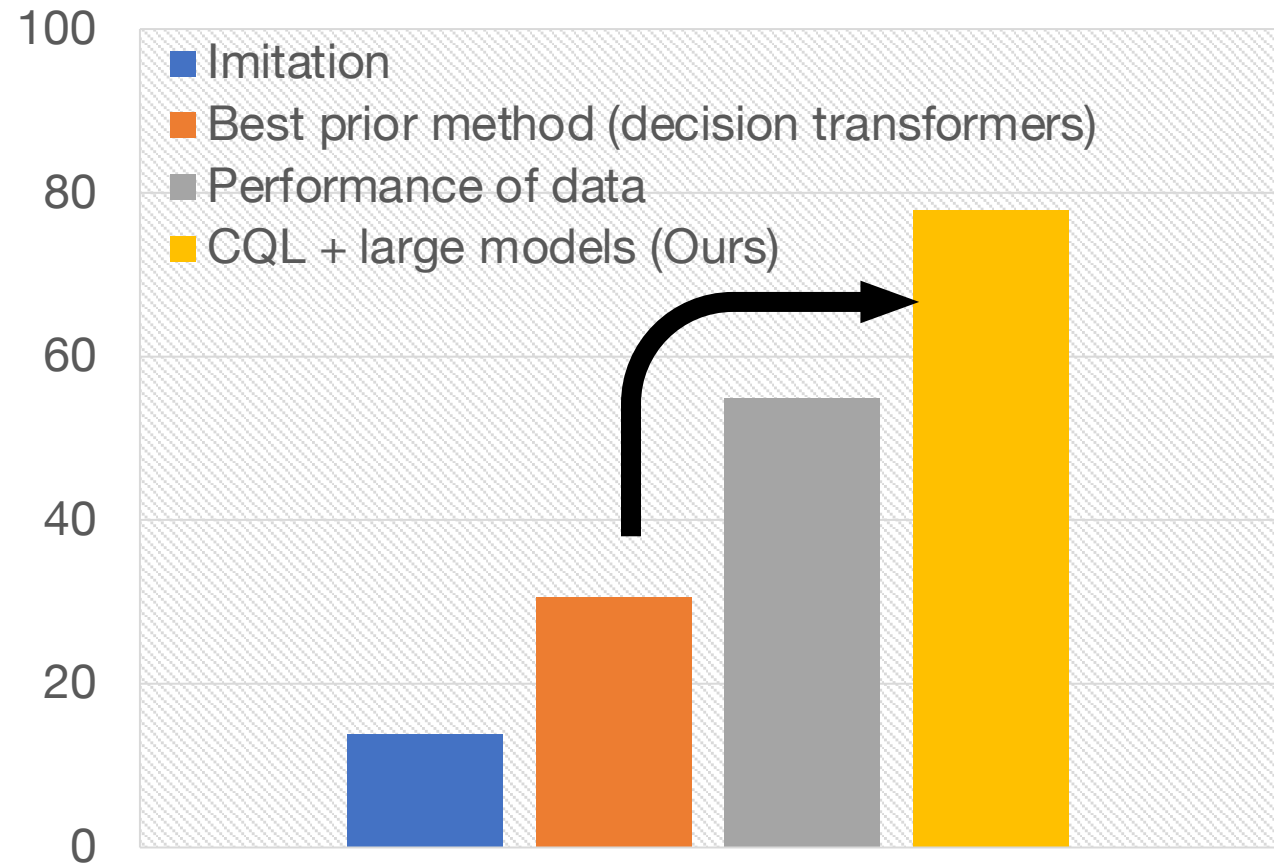
Results: Improves Performance



Train a **single policy** on 40 Atari games

First offline RL approach
to train **large models** to
surpass the data!

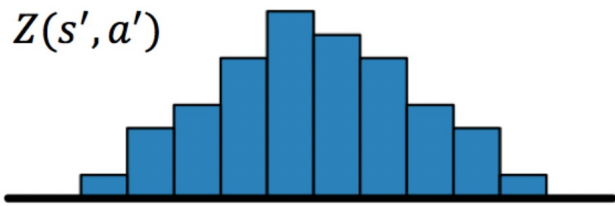
**Human-normalized score
averaged over 40 games**



Can We Push This Further w/ Other Objectives?

Q-value discretization (distributional RL)

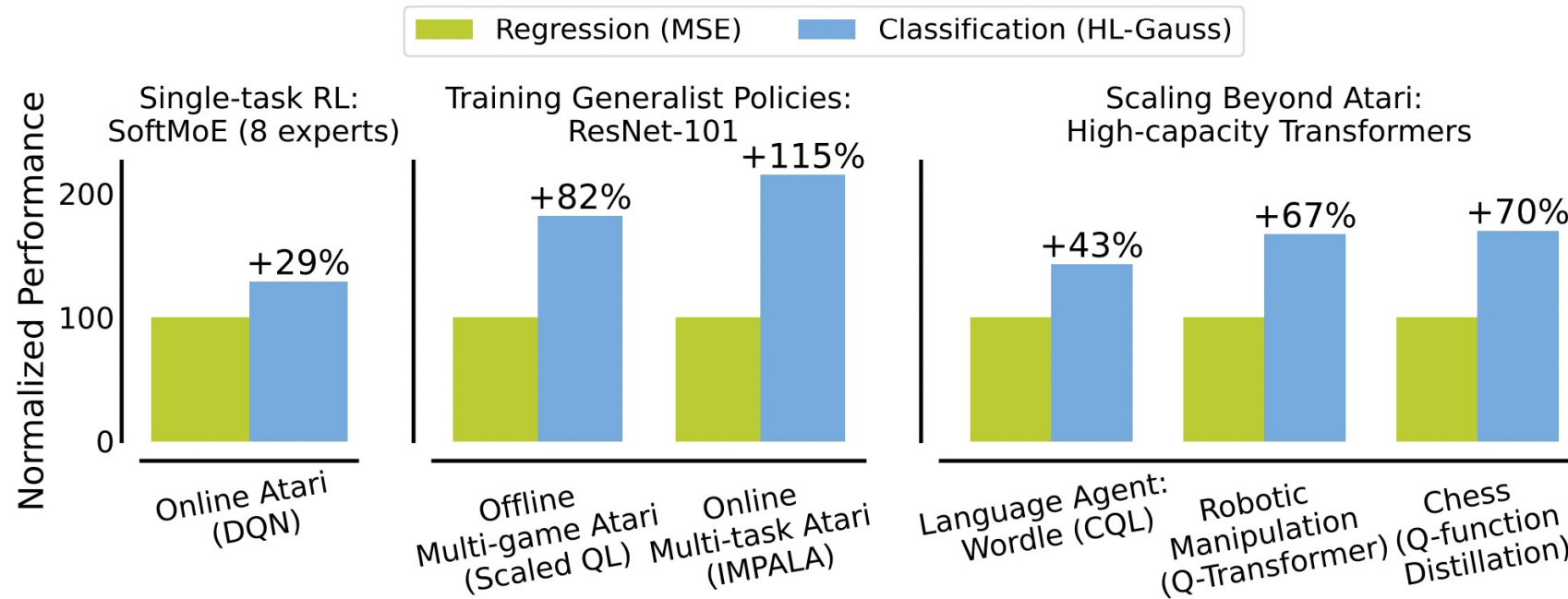
$Z(s', a')$



Q-value = categorical distribution

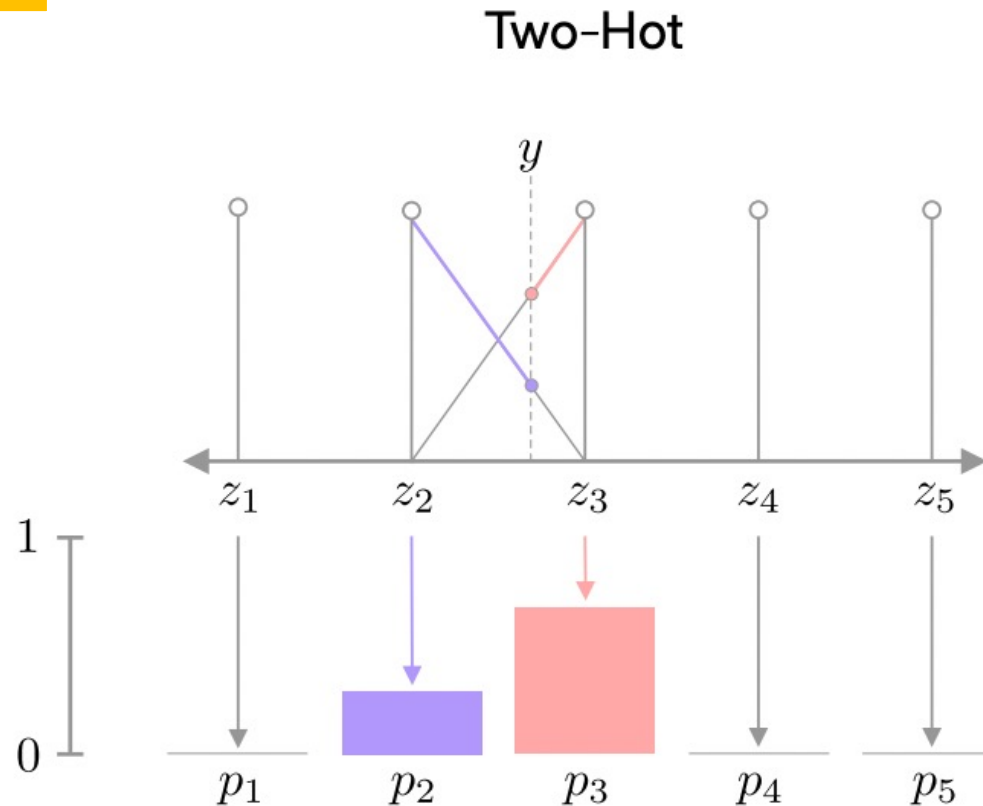
Cross-entropy loss

Distributional RL adds a bias, perhaps the important piece is cross-entropy?



How Can We Design Cross-Entropy Objectives?

The first stab



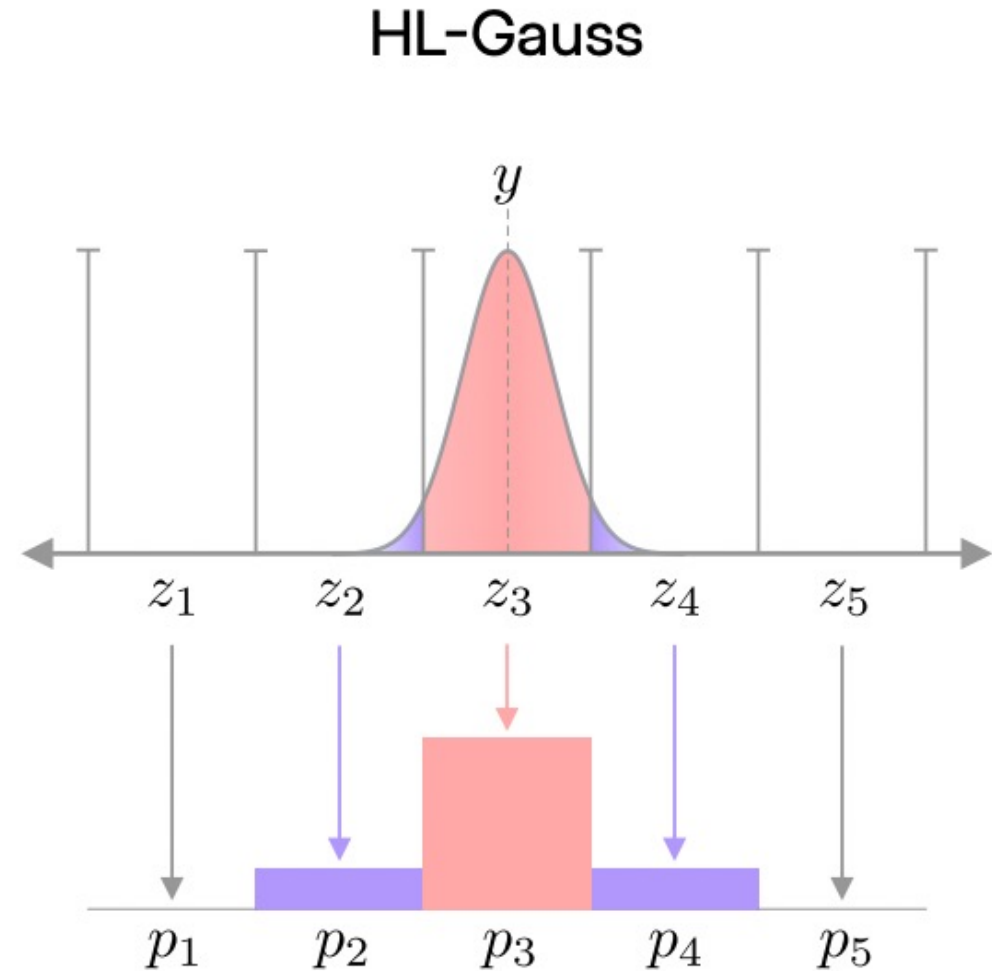
Convert a scalar into probability distribution over two categories

How Can We Design Cross-Entropy Objectives?

A nuanced approach

Two-hot is quite sparse, does not respect ordinal structure

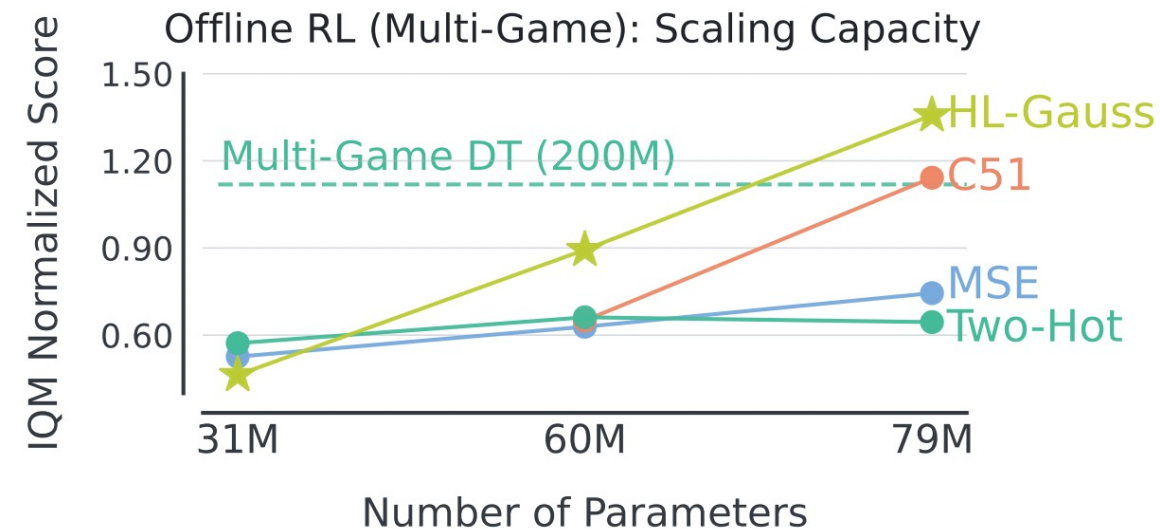
Instead, smoothen out the distribution, add some noise



Very Strong Scaling Results in Several Domains

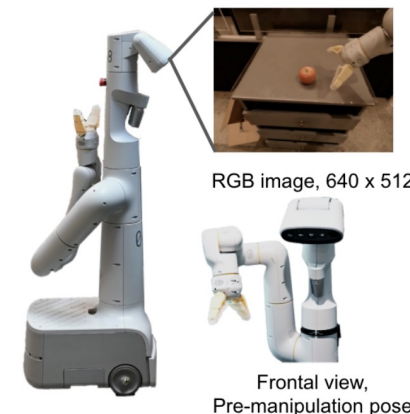
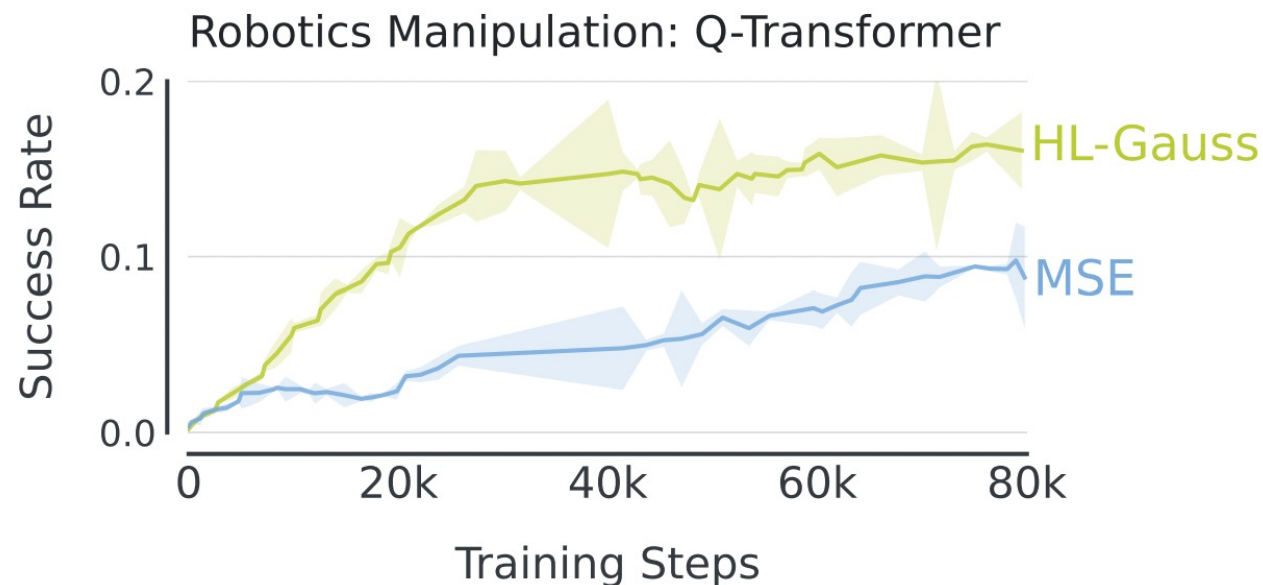
Same multi-game experiment as before; 1.8x better results

Multi-Task Atari



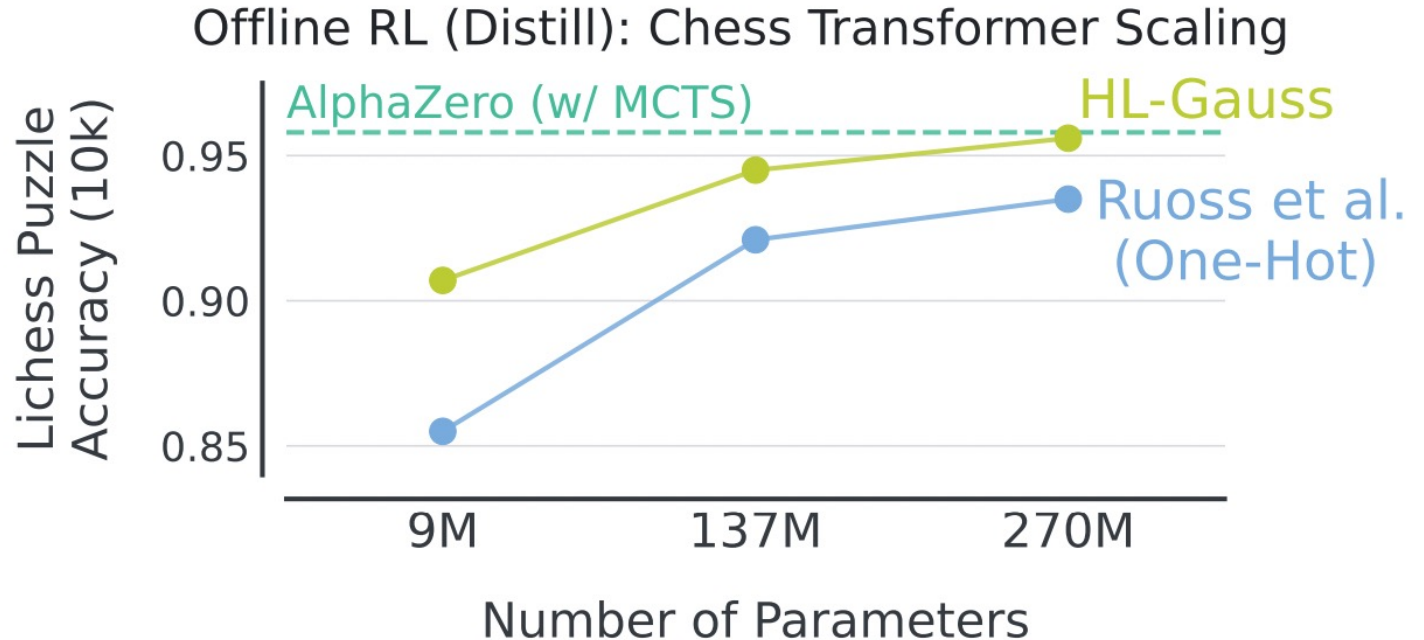
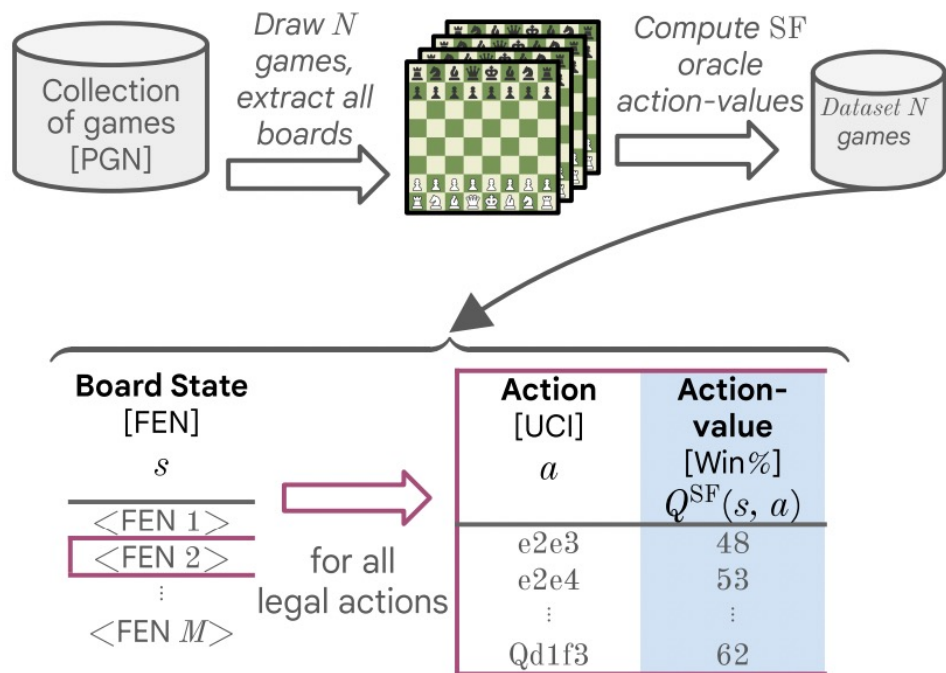
Robot manipulation with tele-op + self-collected data

Multi-Task Robotic Manipulation



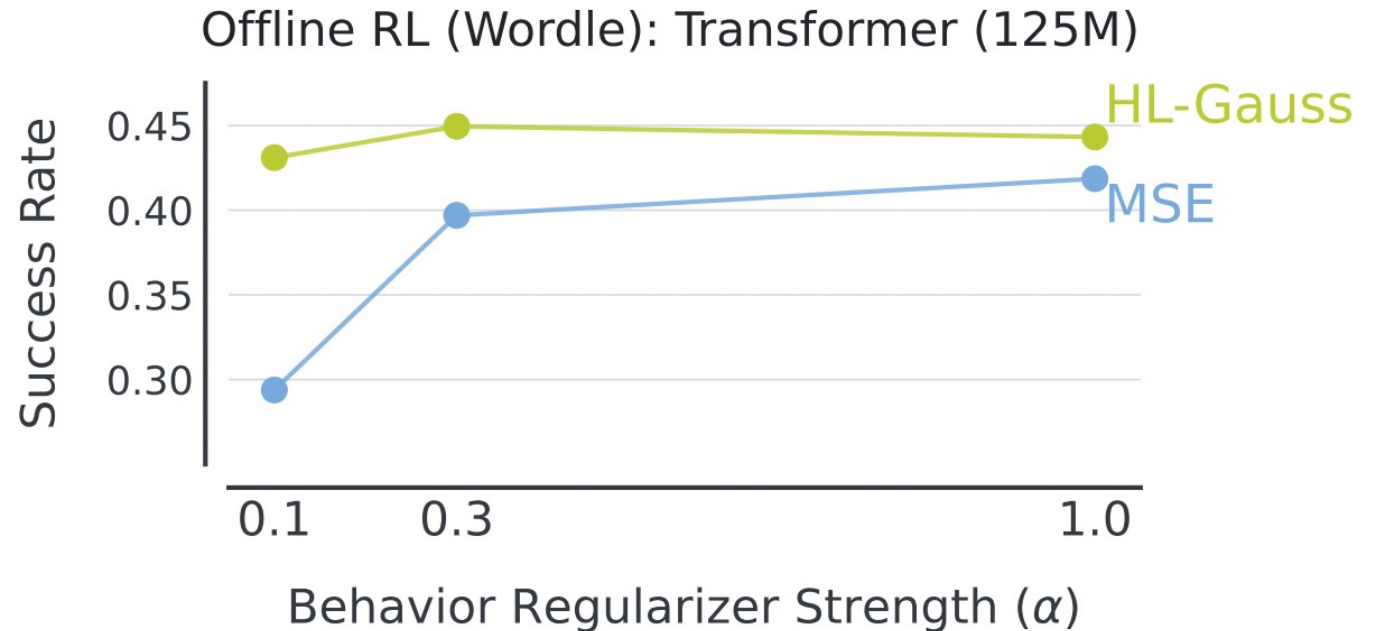
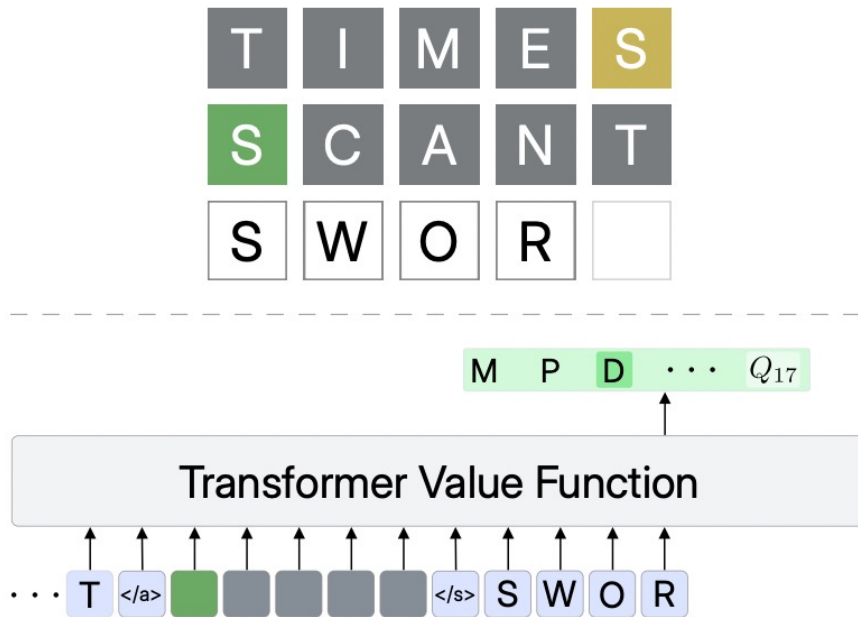
Very Strong Scaling Results in Several Domains

It plays chess at the Grandmaster level, better than AlphaGo Zero (without search)



Very Strong Scaling Results in Several Domains

Also does better and is more robust on LLM Agent Tasks (the game of Wordle)



***Offline RL with
Seq. Models***

Offline + Online RL with Transformers / LLMs

ArCHer: Training Language Model Agents via Hierarchical Multi-Turn RL. Zhou et al. ICML 2024.

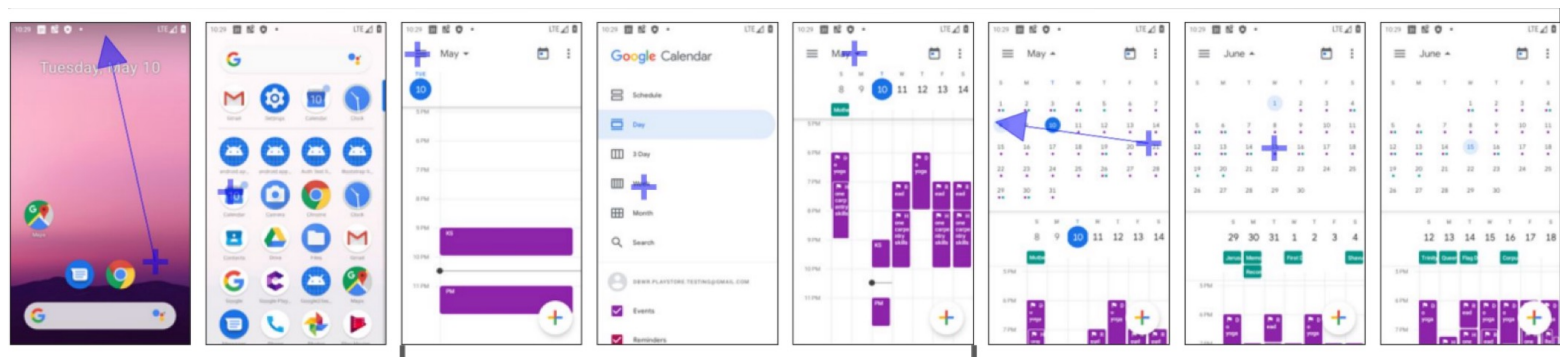
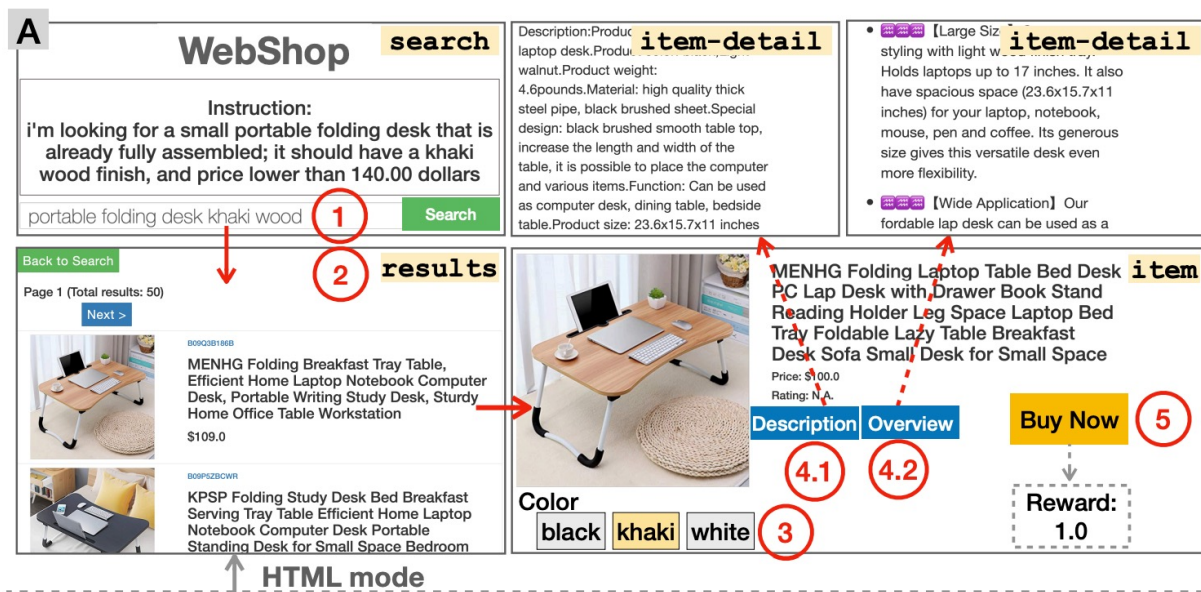
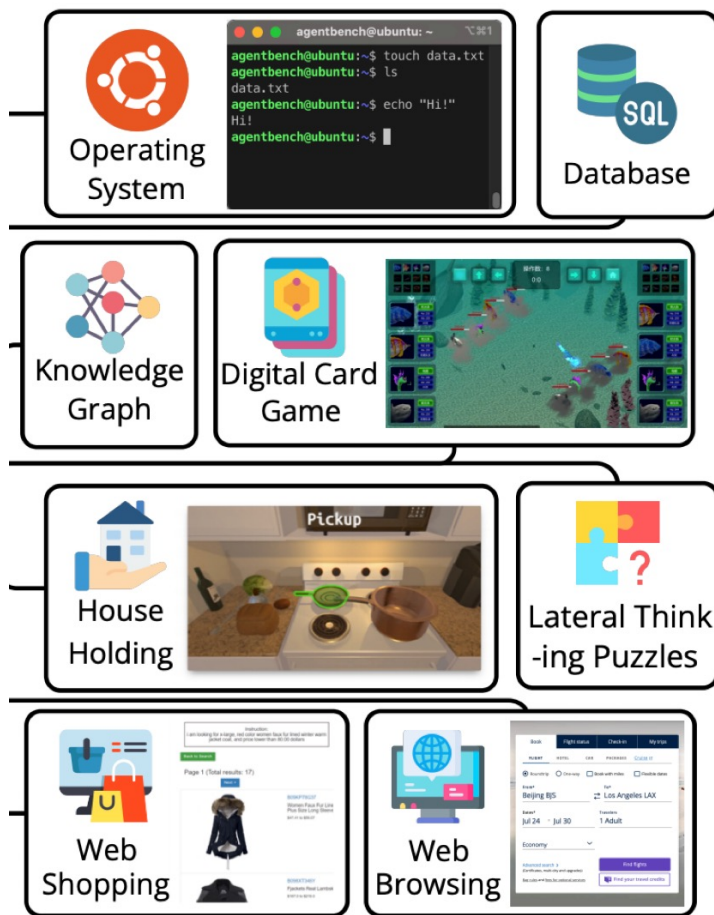
DigiRL: Training In-the-Wild Device Control Agents with Reinforcement Learning. Bai et al. NeurIPS 2024.

Digi-Q: Transforming VLMs to Device-Control Agents via Value-Based Offline RL. Bai et al.

More general algorithms for offline RL:

- Pre-Training for Robots: Offline RL Enables Learning New Tasks in a Handful of Trials. RSS 2023

Using LLMs as Agents and Why It Needs RL...

Liu et al. **AgentBench: Evaluating LLMs as Agents.** arXiv 2024.

Yao et al. **WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents.** NeurIPS 2022

Rawles et al. **Android in the Wild: A Large-Scale Dataset for Android Device Control.** 2023

And To Be Precise, Multi-Turn RL

Prior Solution Strategies in Multi-Turn RL

Desiderata for an approach

- a) Learn from off-policy data
- b) Handle variable-length utterances
- c) Produce a coherent set of tokens

Token-level Q-Learning Methods:
(e.g., Snell et al. 2023; Chebotar et al. 2023)

MDP considers each token as an action



Tractable Bellman backup



Very long horizons
→ signal propagation issues

Turn-Level Q-Learning methods:
(e.g., Verma et al. 2022; all of RLHF)

MDP considers each utterance as an action



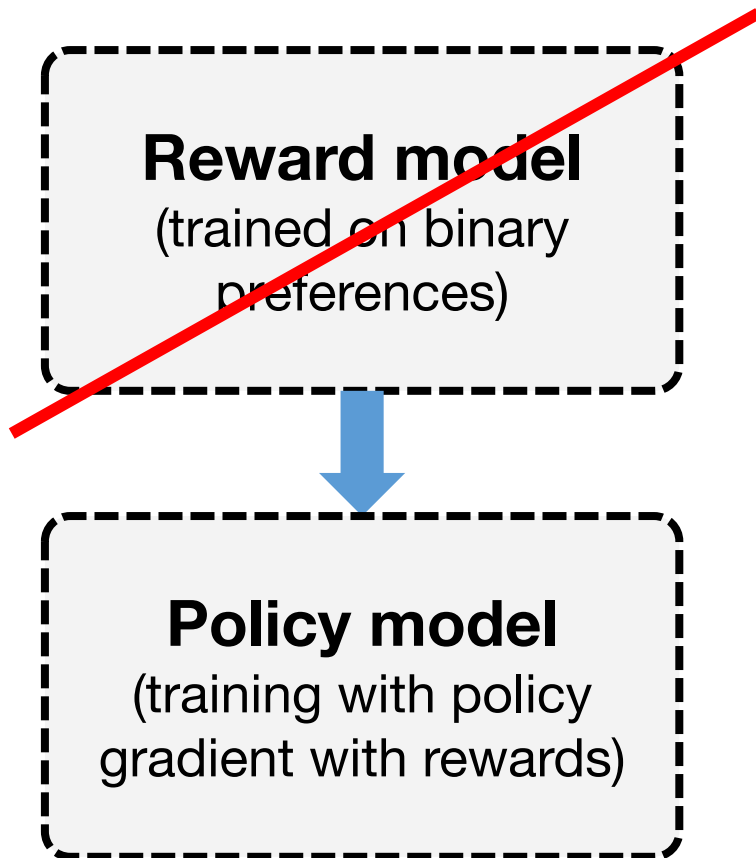
Short horizons



Massively combinatorial
action space

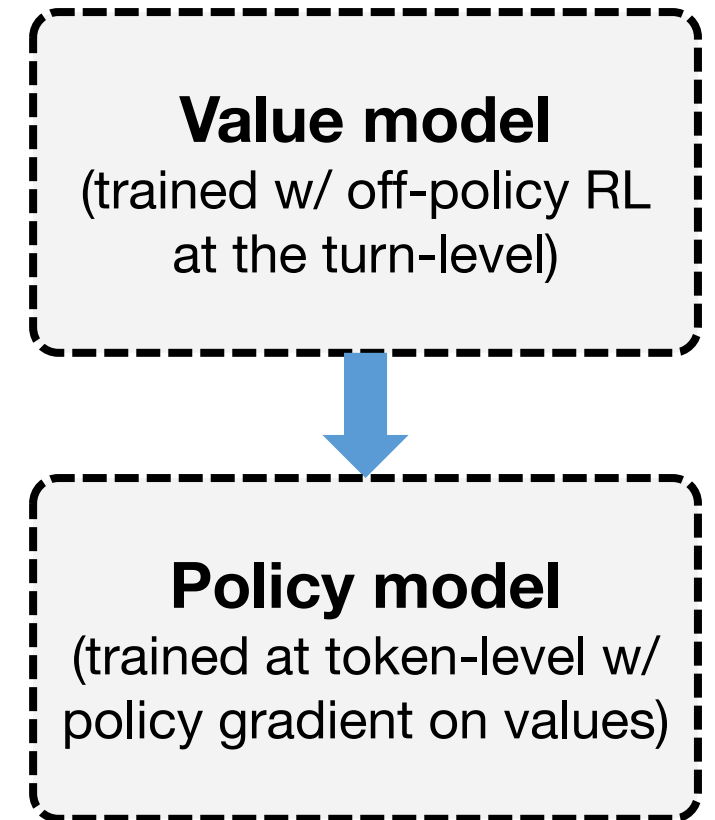
A Starting Point for Multi-Turn RL with LLMs

Single-turn RLHF



Key idea: Replace reward-models with value-models trained at the turn level

Multi-turn RL



ArCHer: Hierarchical Algorithms Generalize This

Multi-turn RL

Value model
(trained w/ off-policy RL
at the turn-level)



Policy model
(trained at token-level w/
policy gradient on values)

Hierarchical Generalization

Can also be over some other
decomposition (not turns)

This can be any “high-level” off-policy
RL algorithm operating on turns

(e.g., Q-learning, value-based RL, model-based RL)

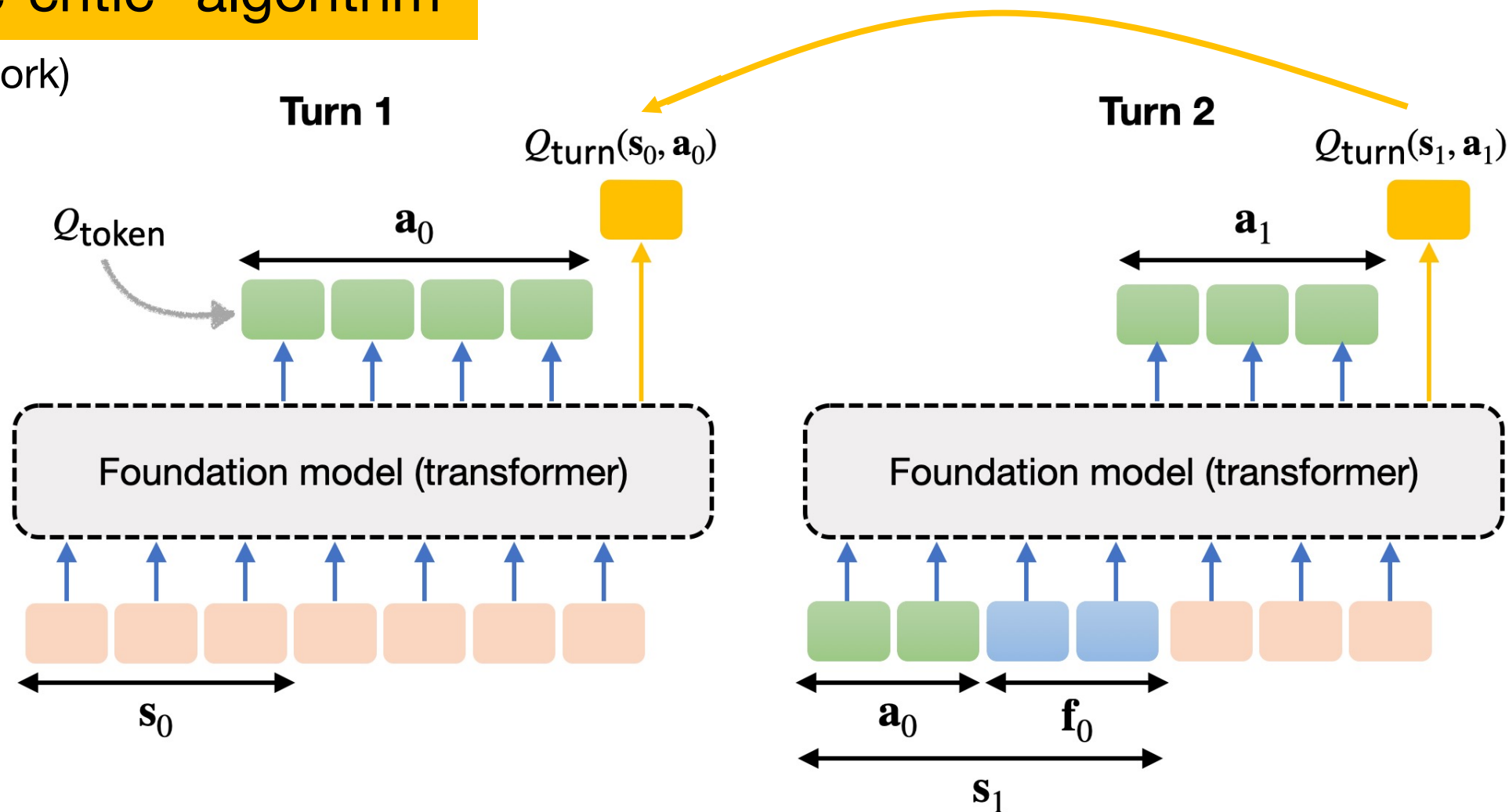
This can be any low-level off-policy /
on-policy RL algorithm on tokens

(e.g., PPO, REINFORCE, Q-learning, etc)

An Example Instantiation with Q-Learning

A “critic-critic” algorithm

(on-going work)



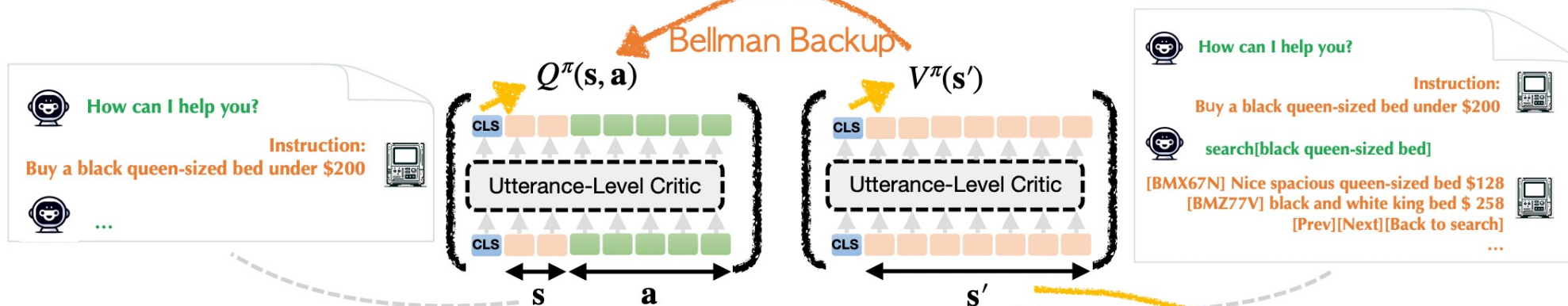
This Paper: An Actor-Critic Practical Algorithm

DQN updates, IQL offline RL updates

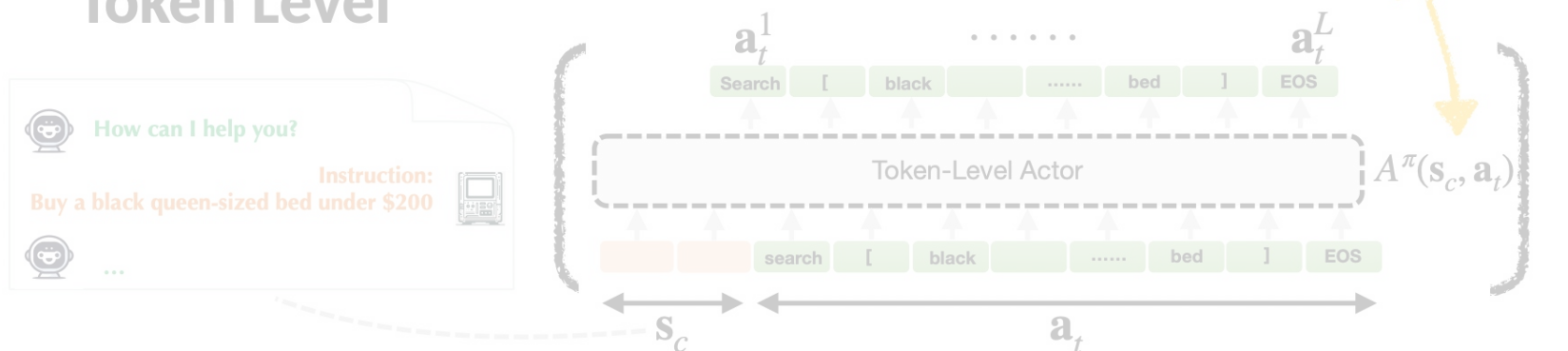
Utterance Level

$$Q^\pi(s, a) \leftarrow r + \gamma V^\pi(s')$$

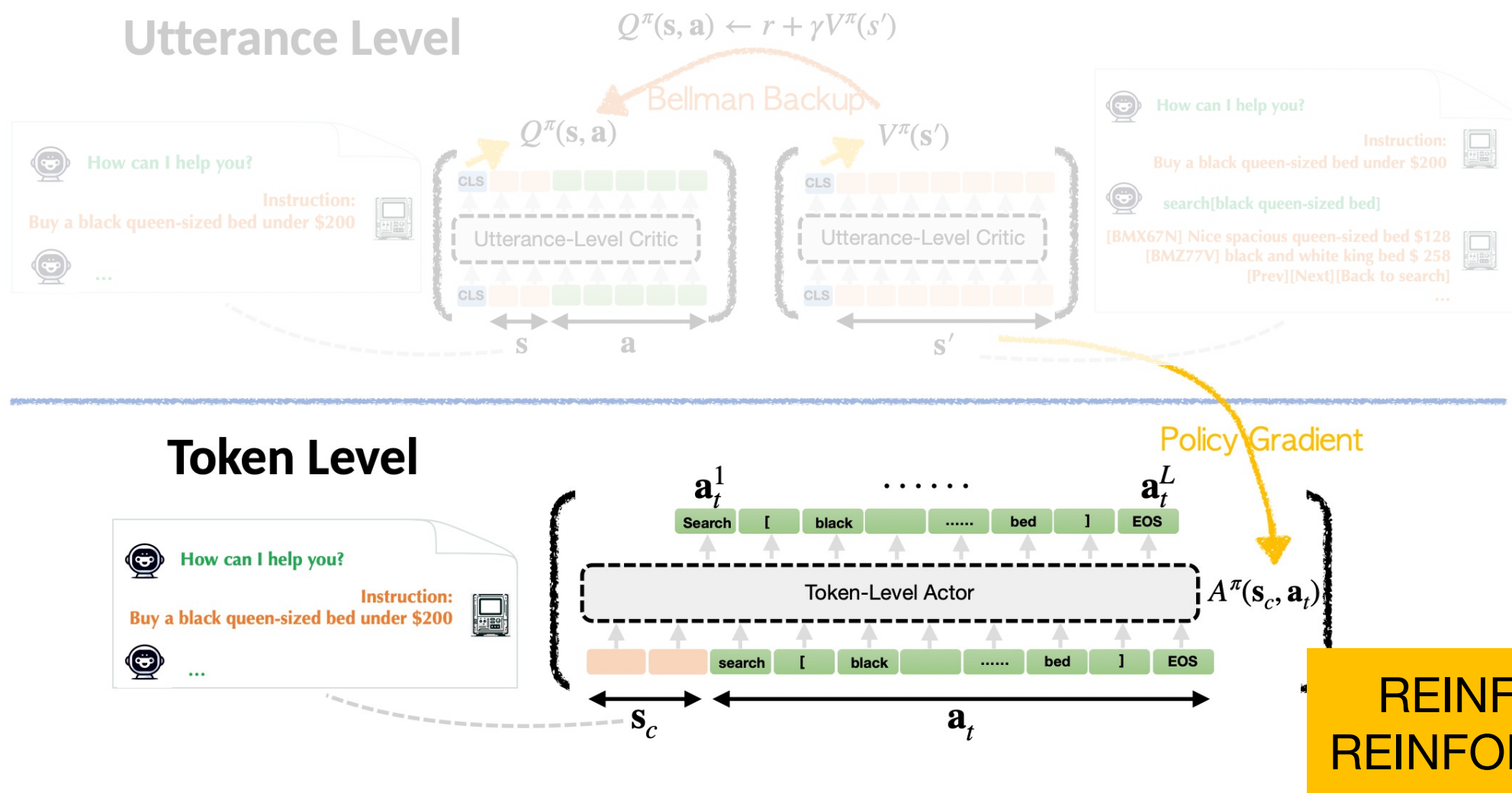
Bellman Backup



Token Level



This Paper: An Actor-Critic Practical Algorithm

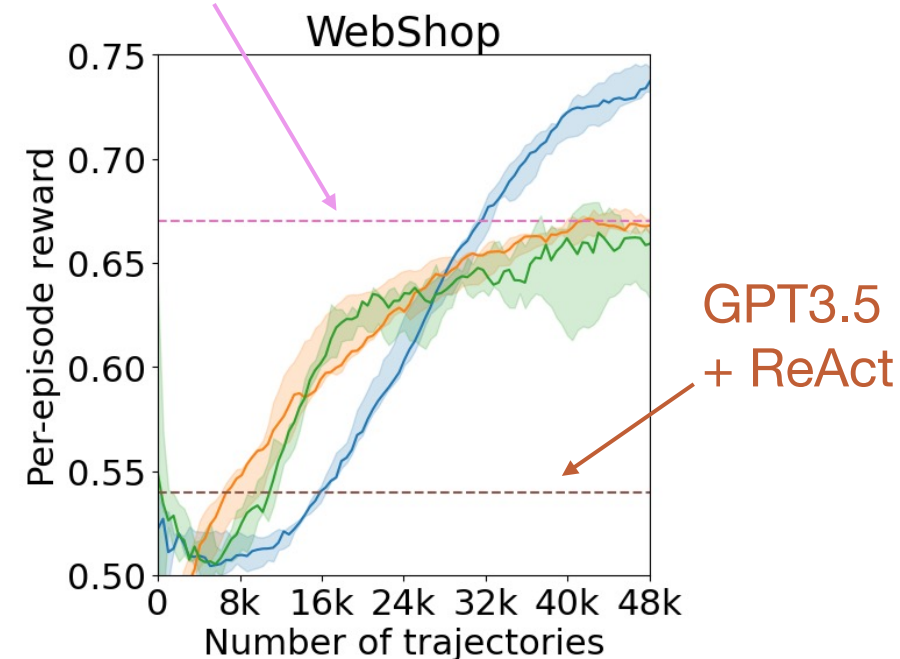
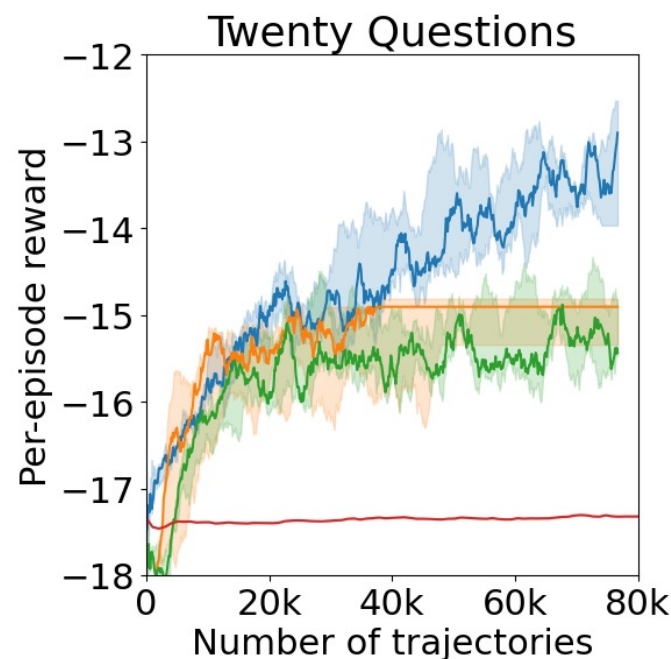
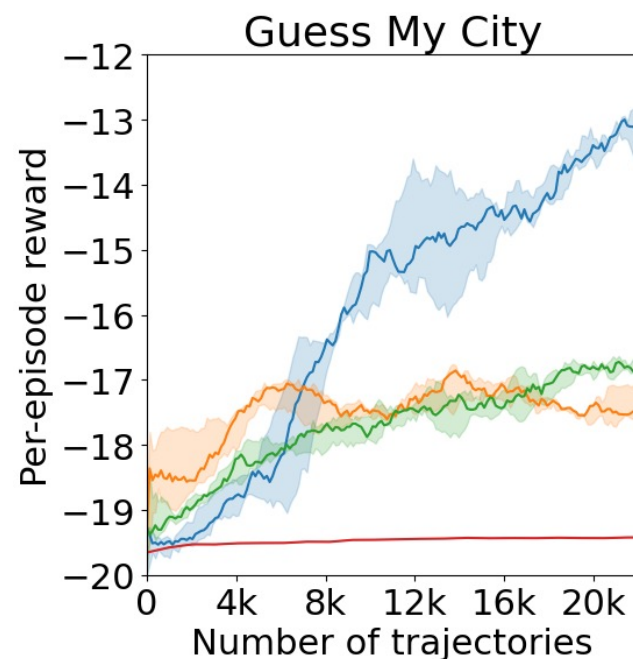


Results: Online Agent Tasks with GPT-2 models

Ask the right questions to identify what the hidden object (e.g., city, hidden word) is

Execute actions on the web

GPT3.5 with expert prompt



ArCher CHAI Filtered BC GPT3.5-expert-prompt GPT3.5-ReAct

Zhou et al. **ArCher: Training Language Model Agents via Hierarchical Multi-Turn RL**. ICML 2024.

Yao et al. **WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents**. NeurIPS 2022.

Abdulhai et al. **LMRL Gym: Benchmarks for Multi-Turn Reinforcement Learning with Language Models**. arXiv 2023.

Example Rollouts in the Environment

Guess My City

Action: Is your city located in Europe?

Observation:

Questions:

What is the primary language spoken in your city?

The primary language spoken in the city is Spanish.

Is your city located on the east coast of the country?

Yes, it is located on the east coast of the country.

Is your city located in North America?

No, my city is not located in North America.

Is your city located in Europe?

Yes, my city is located in Europe.

Twenty Questions

Action: Is the object an artwork?

Observation:

Questions:

Is the object alive?

No.

Is the object man-made?

Yes.

Is the object a vehicle?

No.

Is the object a piece of furniture?

No.

Is the object a piece of clothing?

No.

Is the object a piece of technology?

No.

Is the object a piece of jewelry?

No.

Is the object a piece of clothing?

No.

Is the object an artwork?

No.

Web Shopping

Action:

search[wireless bluetooth 4.0 power amplifier board]

Observation:

WebShop

Instruction:

i am looking for a wireless bluetooth 4.0 power amplifier board, and price lower than 60.00 dollars

[Search]

Action:

search[wireless bluetooth 4.0 power amplifier board]

Observation:

[Back to Search]

Page 1 (Total results: 50)

[Next >]

[B08LDPQCB4]

Audio Amplifier Module,TDA7492P Wireless Bluetooth 4.0 Audio Receiver Power Amplifier Board Module,CSR8635 Bluetooth Chip,Auto Pairing,Amplifier Speakers Board

\$25.19

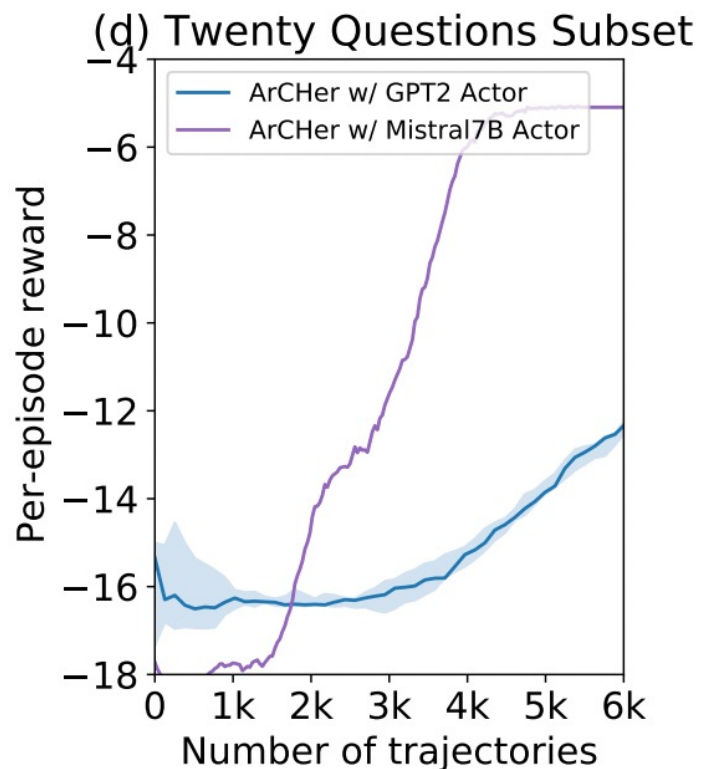
[B08GPVK8LB]

Liccx TDA7492P Power Amplifier Board£-Wireless Bluetooth 4.0 Audio Receiver Power Amplifier Board Module Audio Amplifier Module

\$24.99

Scaling and Algorithm Choices in ArCHer

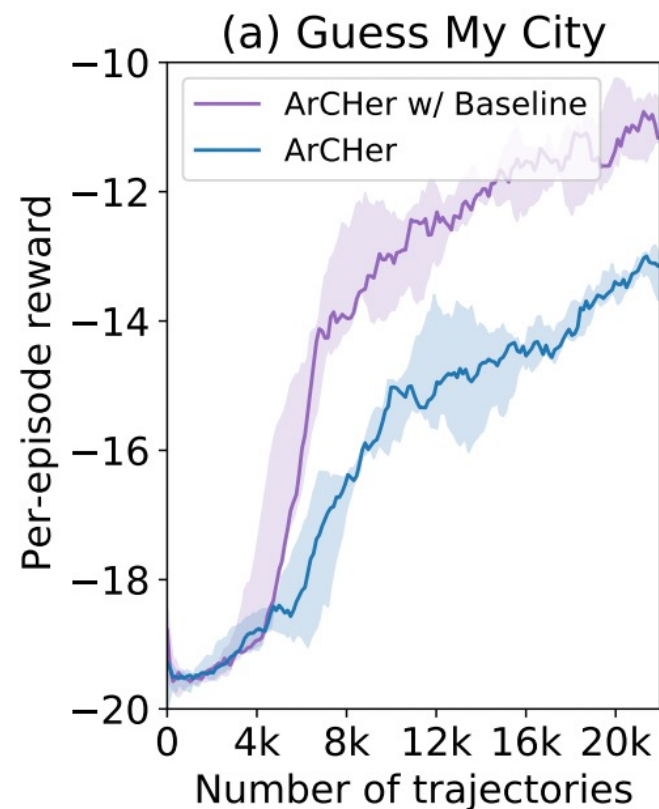
Scaling up to 7B models
(inherits benefits of scaling in RLHF)



Scaling up
Mistral 7B
parameters

My belief:
will scale up
just the same
way as RLHF

Better low-level algos do better
(policy gradient w/ baseline > w/o)



Using a better
low-level
RL algorithm
leads to better
performance

***Offline RL with
Seq. Models***

Takeaways

- **Challenge:** Dealing with big action spaces + long horizons
- **Algorithm:** Hierarchical Multi-turn RL (distribute the challenges into two complementary parts)
- **Result:** Scalable Recipe for Multi-Turn RL Fine-Tuning

***Fine-Tuning
Algorithms***

Online Fine-Tuning of Offline RL

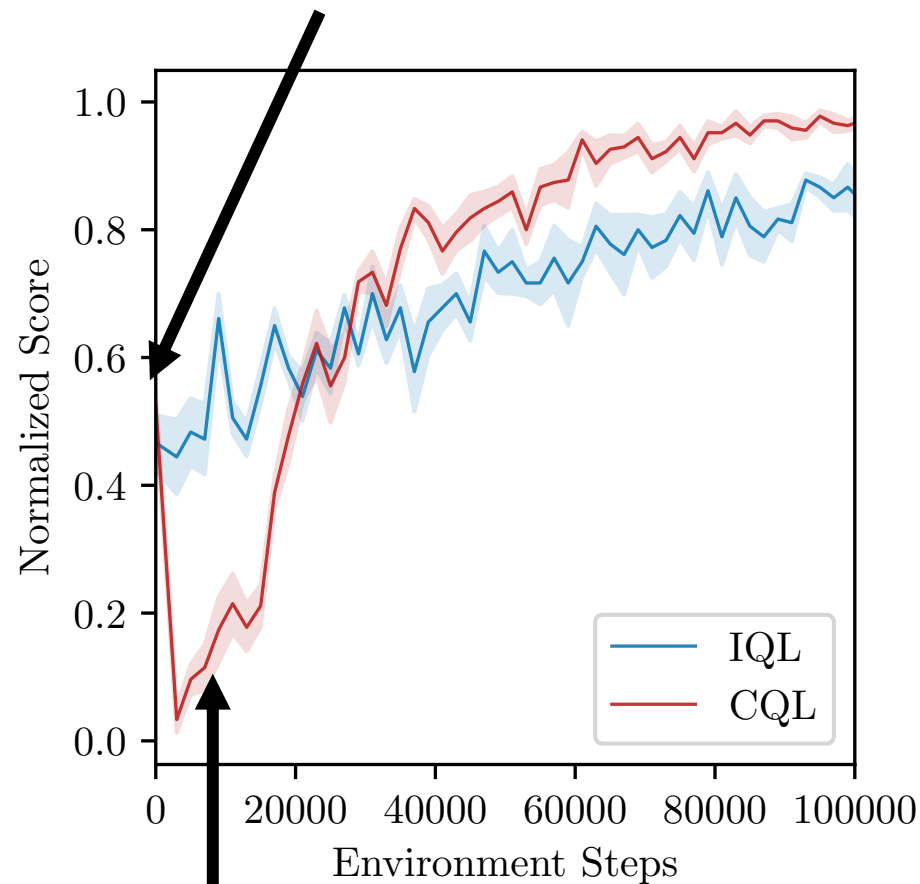
Cal-QL: Calibrated Offline RL Pre-Training for Efficient Online Fine-Tuning.
Nakamoto*, Zhai*, Singh, Mark, Ma, Finn, **Kumar**, Levine. NeurIPS 2023.

Also see:

- Pre-Training for Robots: Offline RL Enables Learning New Tasks in a Handful of Trials. **Kumar** et al. RSS 2023
- Robot Fine-Tuning Made Easy: Pre-Training Rewards and Policies for Autonomous Real-World RL.
Yang*, Mark*, Vu, Sharma, Bohg, Finn. 2023

First Stab: Online RL Fine-Tuning of Offline RL

Start of online data collection + training on it, after offline pre-training



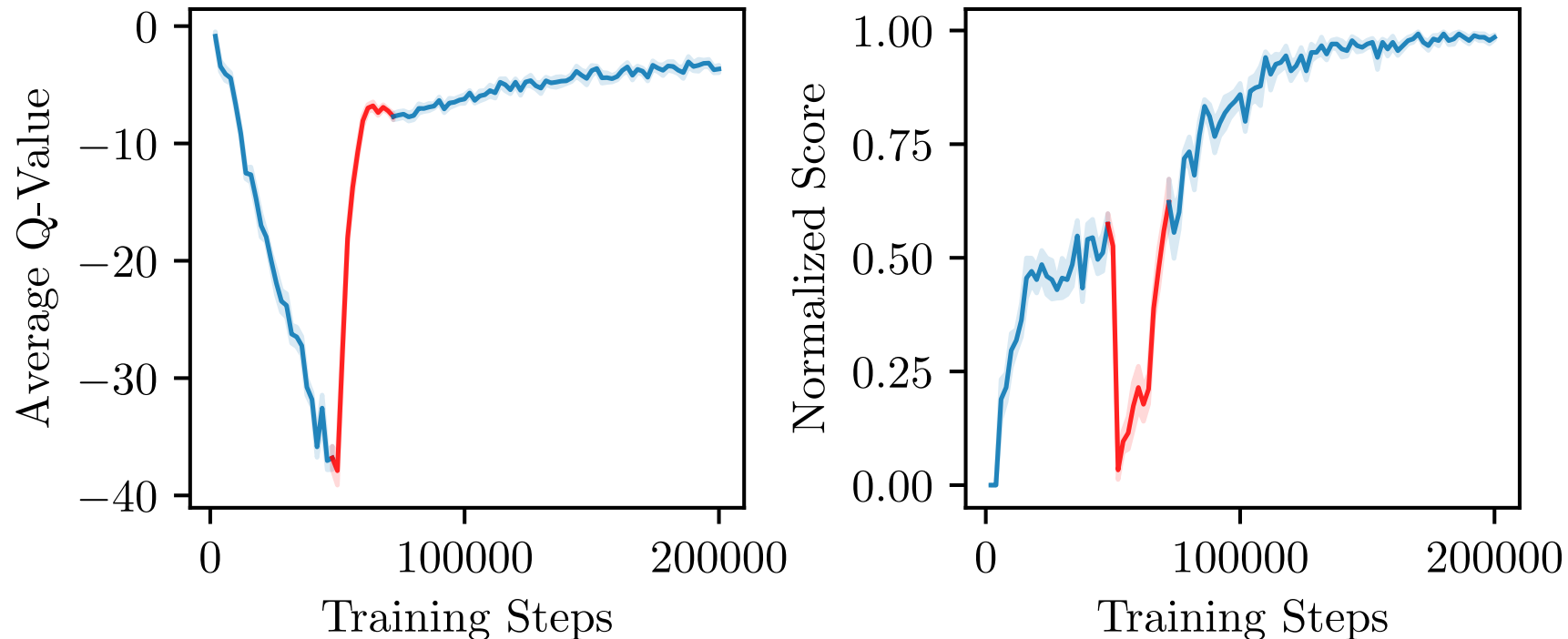
Observations

- Similar offline performance during pre-training
- CQL suffers from **initial policy unlearning**
- IQL improves **slowly** through fine-tuning

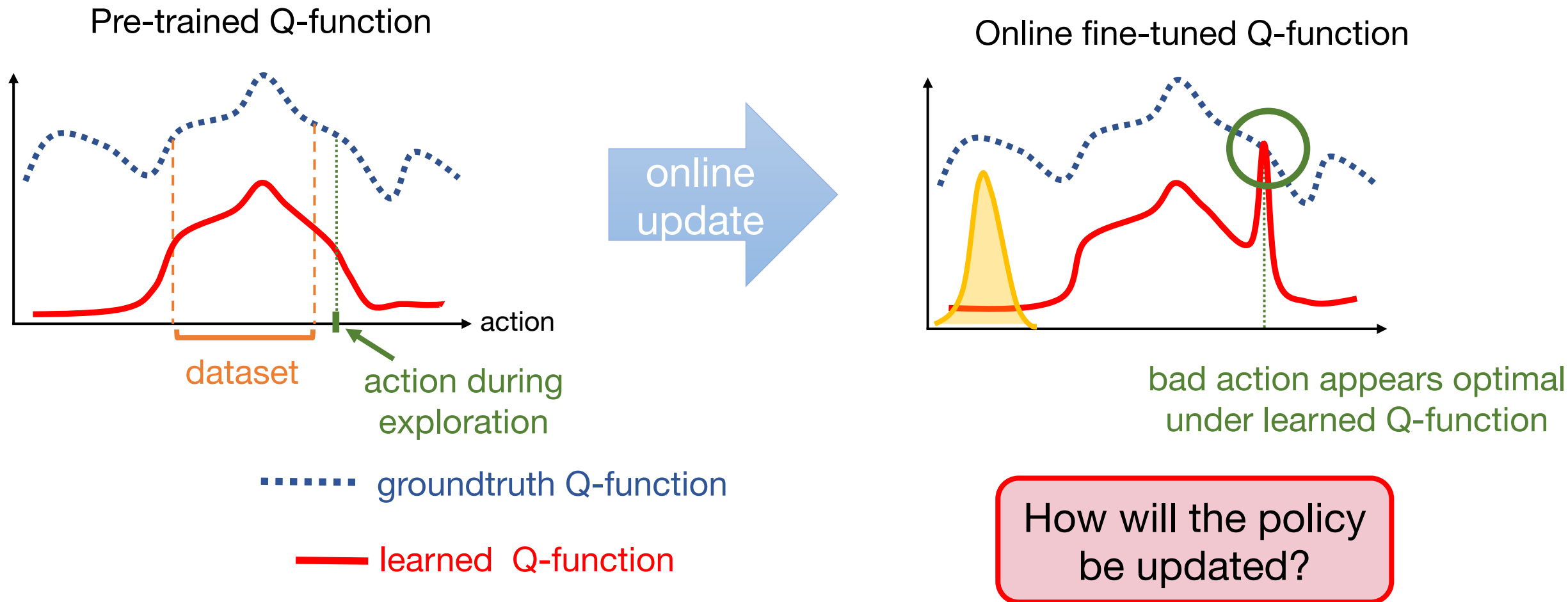
Big drop in performance of CQL right after online fine-tuning begins

Evidence: Why Does CQL “Un-Learn” Initially?

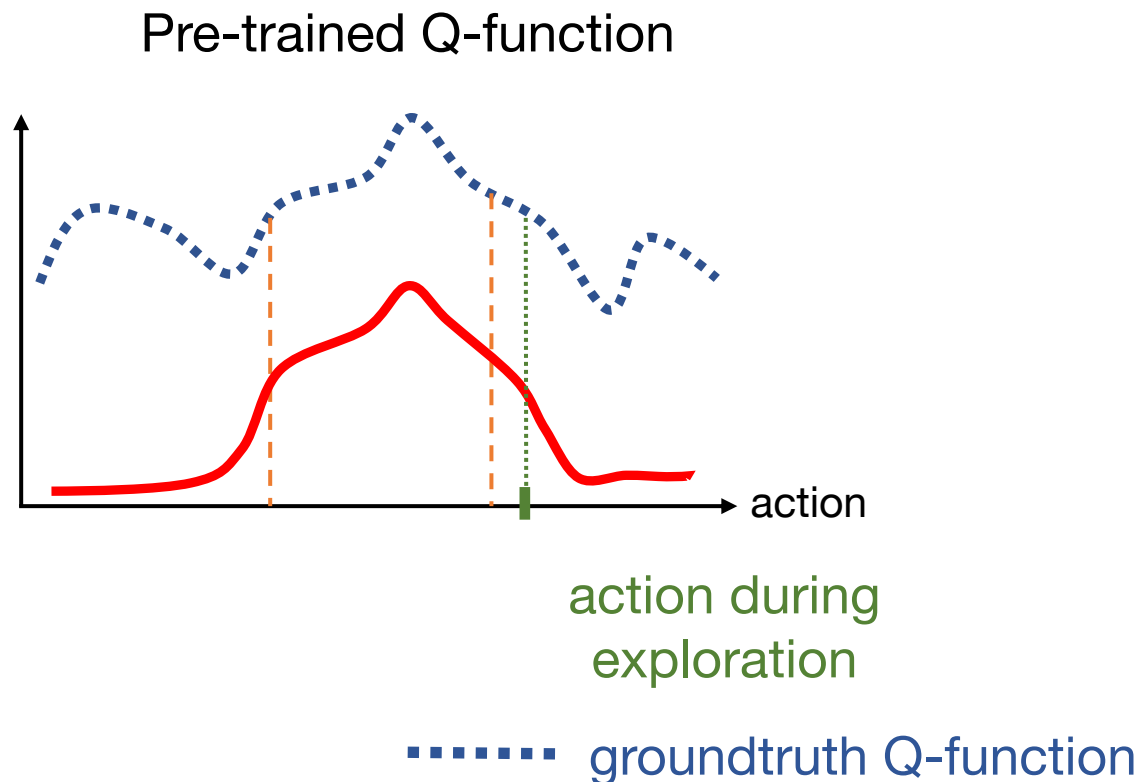
- CQL learns conservative Q values → magnitudes are much smaller
- But these Q-values drastically adjust in scale when online fine-tuning begins
- Online samples wasted, before online RL can run fine.



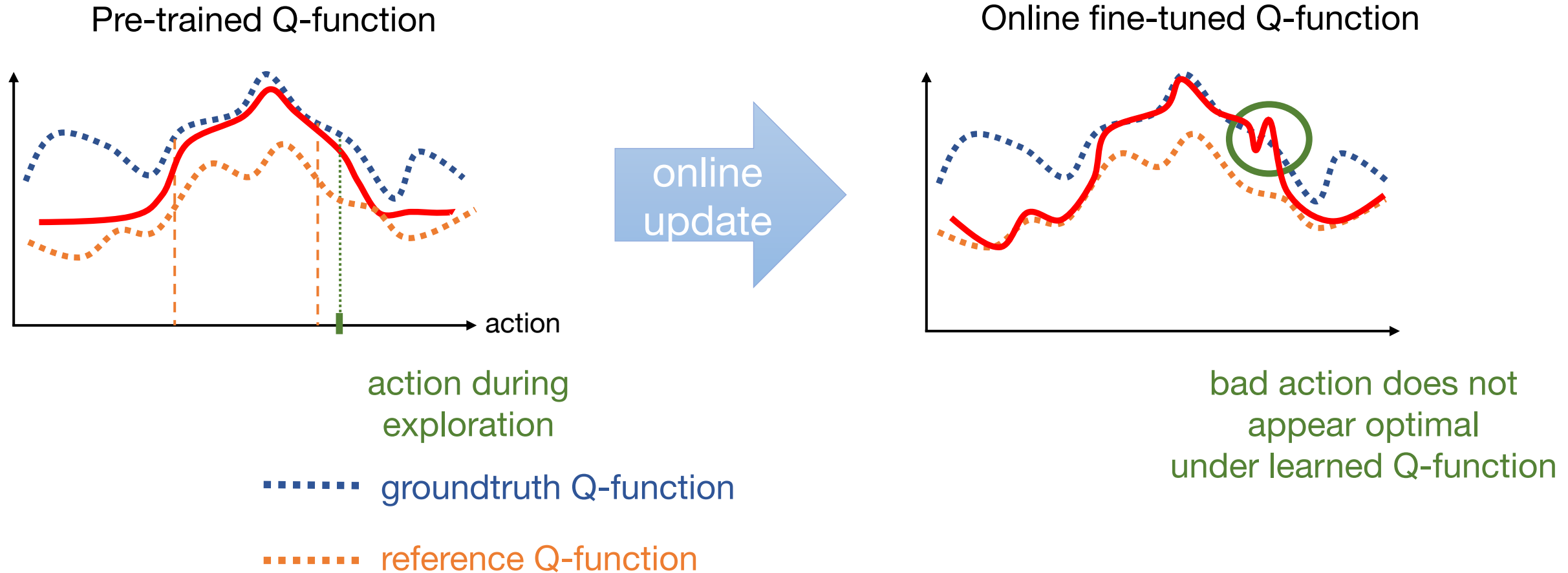
Intuition: Why Does CQL “Un-Learn” Initially?



Calibrating Conservative Q-Functions



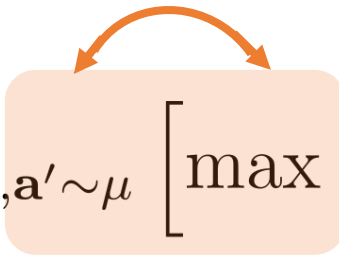
Calibrating Conservative Q-Functions



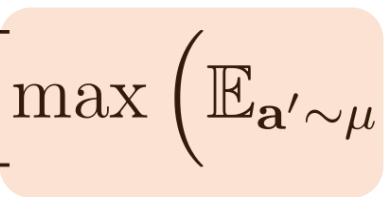
Summary: Calibrated Q-Learning (Cal-QL)

$$\min_{\theta} \max_{\mu} \alpha \left(\mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{a}' \sim \mu} \left[\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}') \right] - \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} \left[\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}) \right] \right) + \text{TD-Error}(\theta)$$

Intuition: Never push down Q-values if they are smaller than a **reference Q-value**

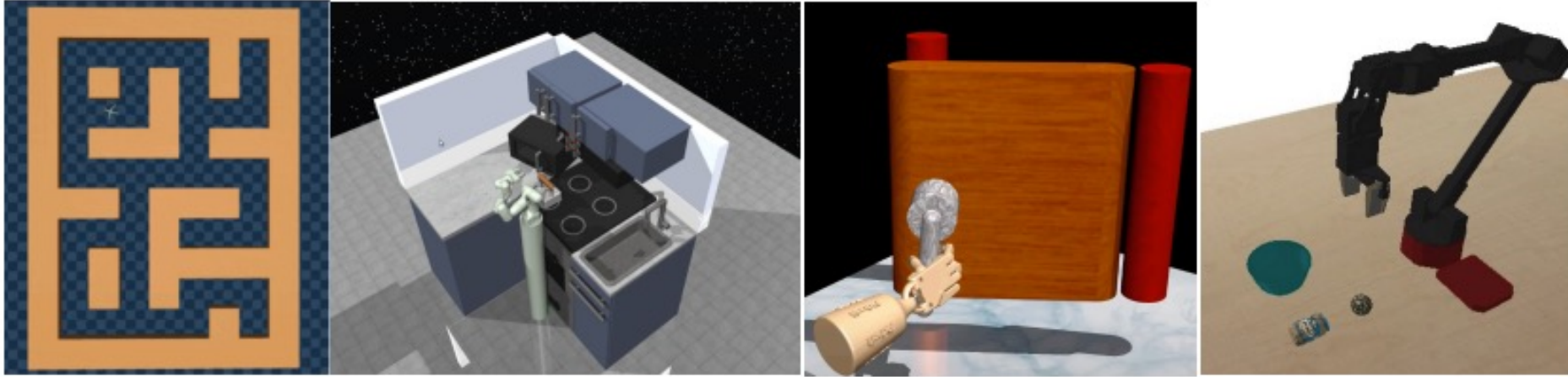

$$\min_{\theta} \max_{\mu} \alpha \left(\mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{a}' \sim \mu} \left[\max \left(\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}'), Q^{\mu}(\mathbf{x}, \mathbf{a}') \right) \right] - \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} \left[\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}) \right] \right) +$$

Blocks out the gradient if the Q-value is worse than reference policy


$$\min_{\theta} \max_{\mu} \alpha \left(\mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\max \left(\mathbb{E}_{\mathbf{a}' \sim \mu} [\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}')] , V^{\mu}(\mathbf{x}) \right) \right] - \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} \left[\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}) \right] \right) +$$

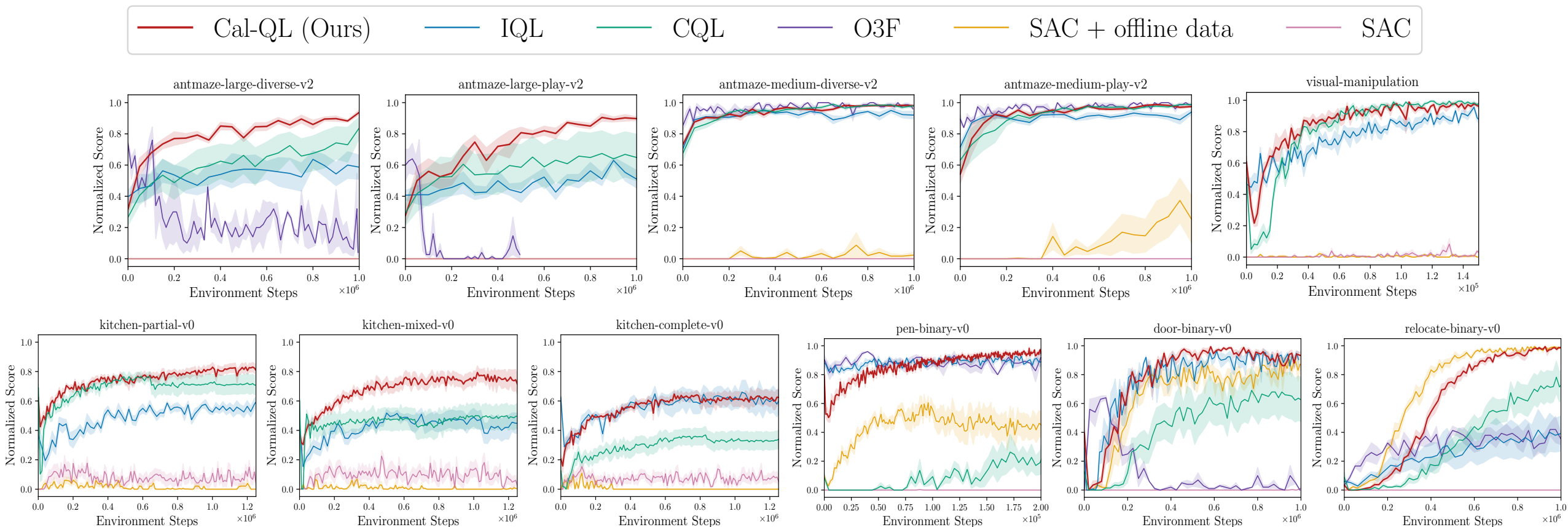
Ultimately care about the expected value function

Benchmark Results for Cal-QL



- **“Stitching” tasks:** AntMaze and FrankaKitchen tasks [**Deviate away from sub-optimal data**]
- **Dexterous manipulation:** Adroit hand tasks [**offline data is expert, but must refine the policy**]
- **From raw observations:** Visual robotic manipulation tasks [**requires sample-efficient learning**]

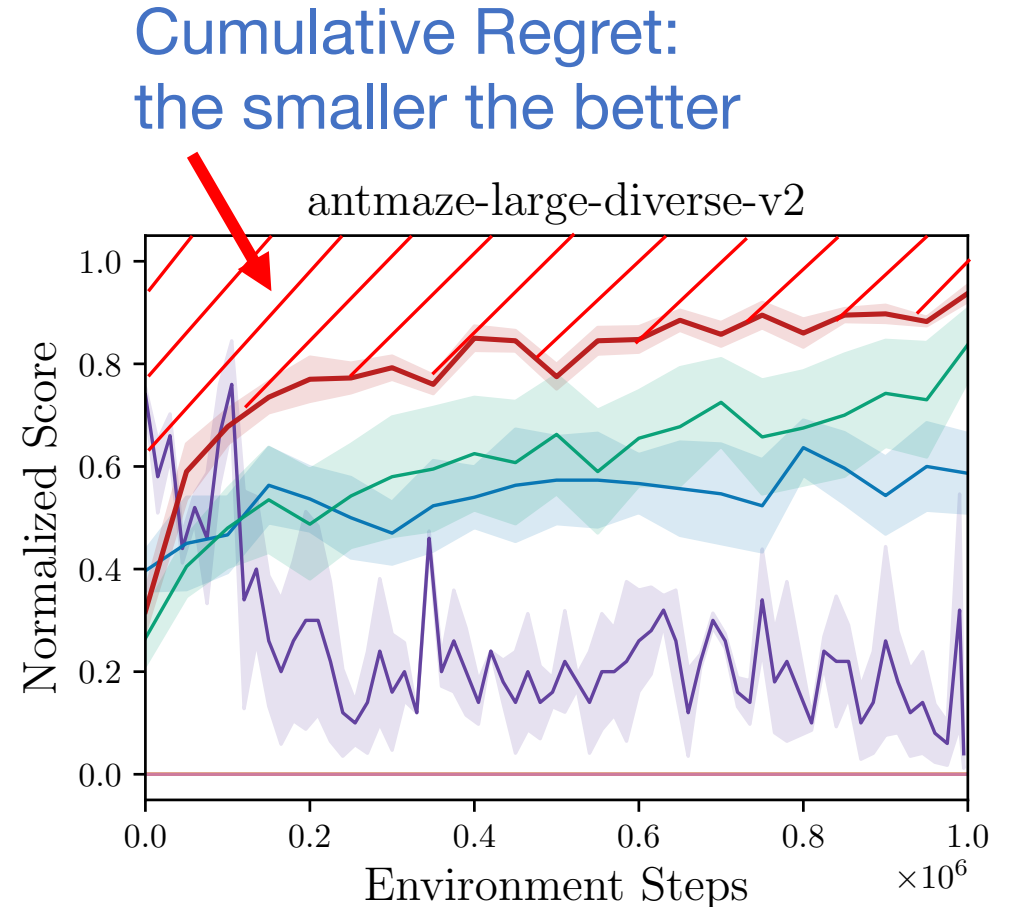
Cal-QL Outperforms Prior Methods



Using a higher **UTD** ratio (=20) for online fine-tuning can further improve the sample efficiency.

Cal-QL Attains Smallest Cumulative Regret

Task	IQL	CQL	SAC+od	SAC	O3F	Cal-QL
large-diverse	0.46	0.39	1.00	1.00	0.75	0.21
large-play	0.52	0.43	1.00	1.00	0.88	0.27
medium-diverse	0.09	0.06	0.98	1.00	0.03	0.06
medium-play	0.10	0.08	0.90	1.00	0.04	0.08
partial	0.51	0.32	0.99	0.92	N/A	0.26
mixed	0.58	0.54	0.99	0.91	N/A	0.32
complete	0.44	0.71	0.99	0.93	N/A	0.44
pen-binary	0.12	0.92	0.56	1.00	0.12	0.12
door-binary	0.24	0.53	0.30	1.00	0.87	0.17
relocate-binary	0.75	0.65	0.20	0.75	0.69	0.31
visual-manipulation	0.24	0.19	1.00	0.99	N/A	0.15
average	0.37	0.44	0.81	0.95	0.48	0.22



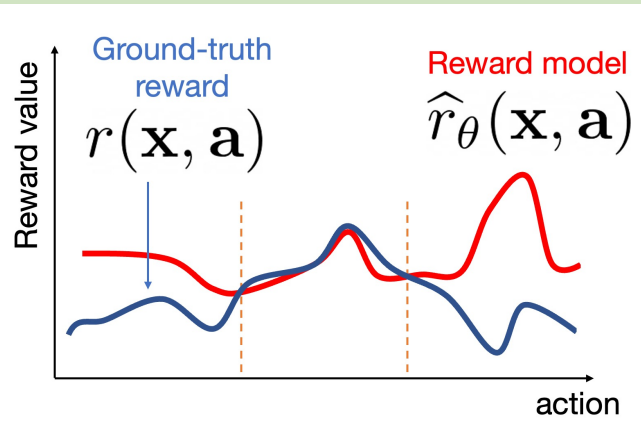
Also see: (Cal-QL with VLM rewards)

Robot Fine-Tuning Made Easy: Pre-Training Rewards and Policies for Autonomous Real-World RL.

Yang*, Mark*, Vu, Sharma, Bohg, Finn. 2023

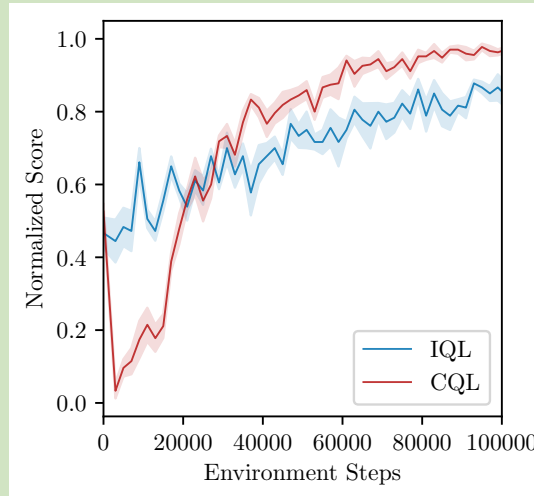
Today's Lecture: Several Facets of Offline RL

Algorithms for Offline RL



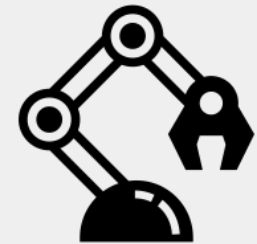
Conservative Q-learning,
Model-based offline RL,
large-scale results

Extensions of Offline RL



Online fine-tuning,
offline RL with LLMs, etc

Applications in Robotics

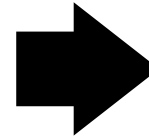


Robotic pre-training + fine-tuning,
learning from videos


$$|\tau| \leq 1 - |\tau|$$

2. Continue offline / online fine-tuning with RL

Task: Solving A Task in A New Domain



2. Fine-Tune on Target Domain Data:
1 door, 10 demonstrations

1. Pre-Train on Bridge Data, 12 doors 800 demonstrations

Results: Solving A Task in A New Domain

Method:

Imitation (Best prior method)

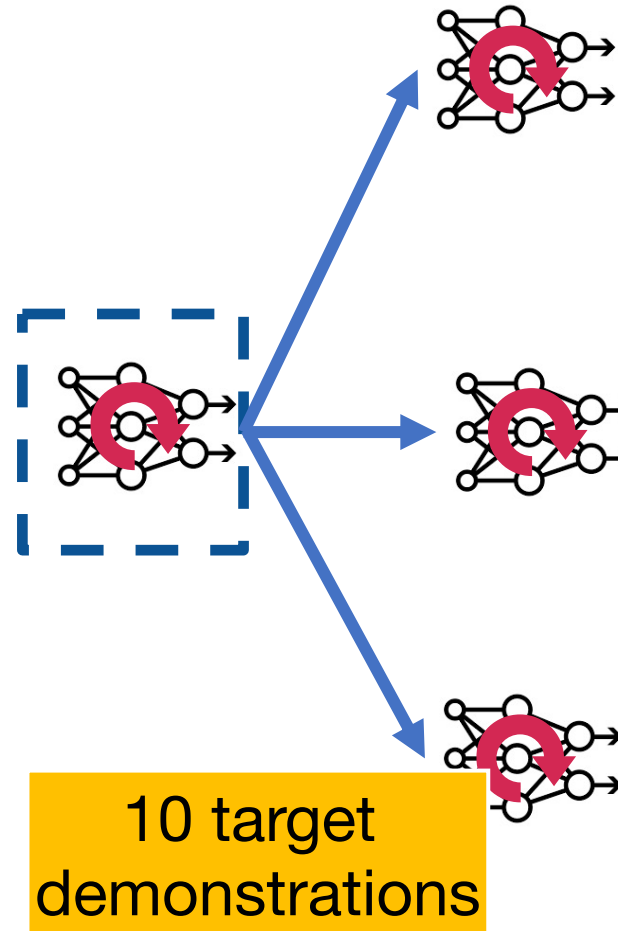


Method:

PTR (Ours)



Task: Solving **New** Tasks in New Domains



Results: Solving **New** Tasks in New Domains

**Best Prior
method**



PTR (Ours)



Some Quantitative Results

Joint training vs pre-training??

Task	PTR (Ours)	BC finetuning			Joint training		Target data only		Pre-train. rep. + BC finetune	
		BC (fine.)	Autoreg. BC	BeT	COG	BC	CQL	BC	R3M	MAE
Take croissant from metal bowl	7/10	3/10	5/10	1/10	4/10	4/10	0/10	1/10	1/10	3/10
Put sweet potato on plate	7/20	1/20	1/20	0/20	0/20	0/20	0/20	0/20	0/20	1/20
Place knife in pot	4/10	2/10	2/10	0/10	1/10	3/10	3/10	0/10	0/10	0/10
Put cucumber in pot	5/10	0/10	1/10	0/10	2/10	1/10	0/10	0/10	0/10	0/10

Imitation (using transformers,
auto-regressive)



Better fine-tuning!

Self-supervised pre-training
from videos / bridge data



Representation learning

Takeaway: Offline RL learns useful representations + better fine-tuning

Analysis

Why would RL enable better performance...

....when the data is collected via human teleoperation?

Preview: Value-functions
can learn what's critical!

Qualitative Comparison of BC (finetune) and PTR

Task: Take Croissant from Metal Bowl



BC (finetune)
Failure: grasps bowl instead of croissant when
croissant is not underneath



PTR
Success: grasps croissant and puts by sink

Task: Put Cucumber in Bowl

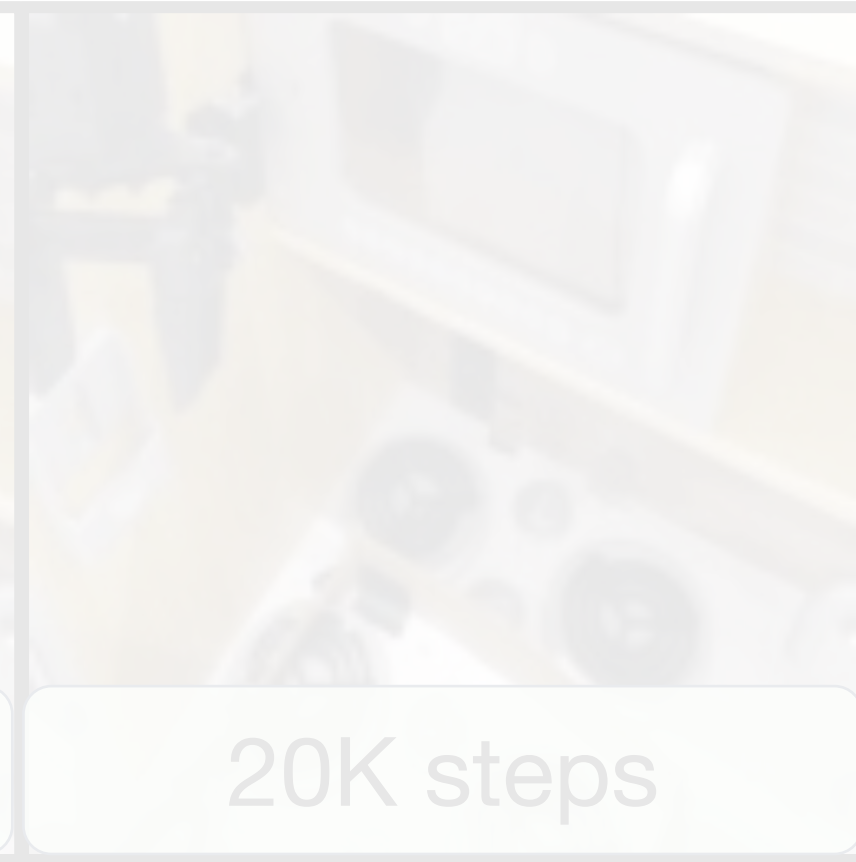
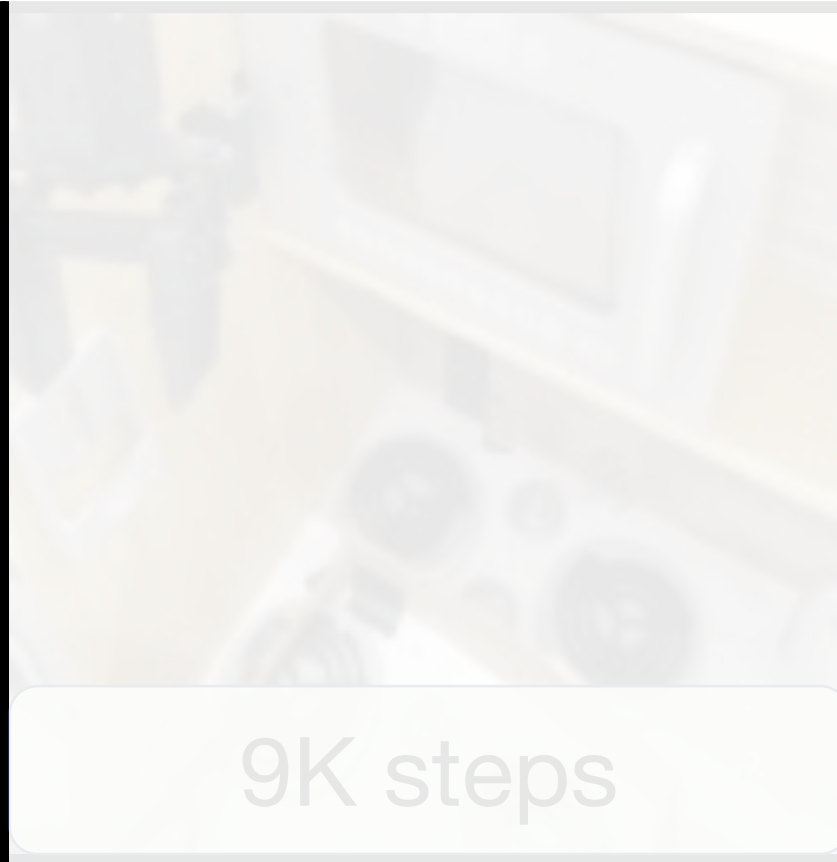


BC (finetune)
Failure: executes an imprecise grasp, and fails to
locate the pot accurately



PTR
Success: Places Cucumber in Pot

Autonomous Improvement via Cal-QL



Autonomous Improvement via Cal-QL



Initialization



9K steps



20K steps

Autonomous Improvement via Cal-QL



Initialization



9K steps



20K steps

Application: Using Human Videos via Offline RL

Internet-Scale Human Video



Task Specific Data

Multi-task Robot Data

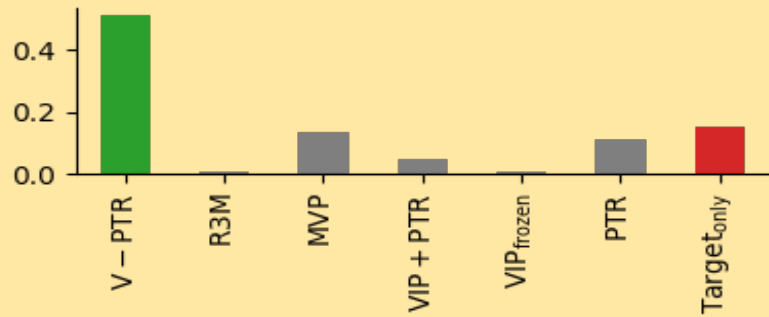
Human video data (e.g., Ego4D)
from the internet

Challenge: No actions in video
data, different embodiment.

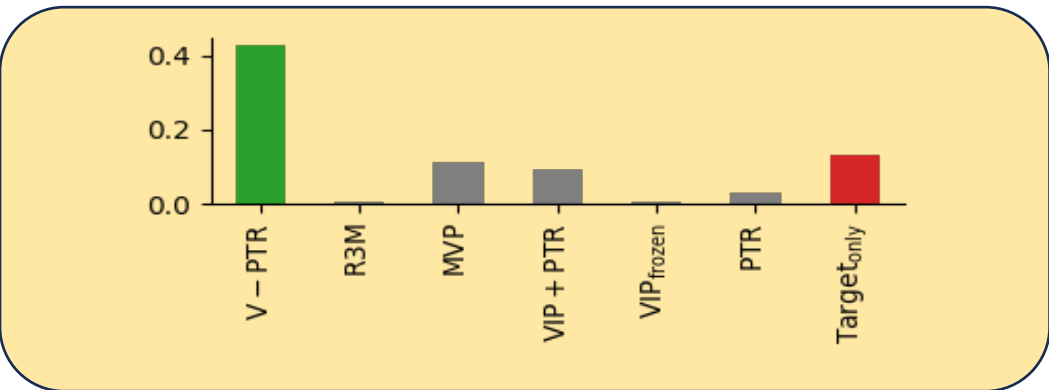


Can still learn value functions
 $V(s)$ and use representations

Scenario 1: Object and Gripper Variability

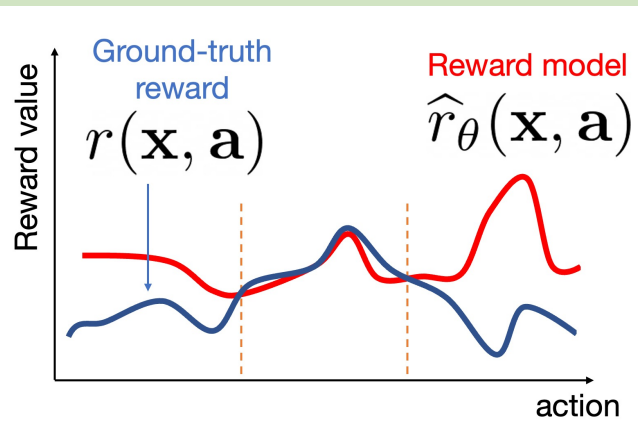


Scenario 2: Distractors



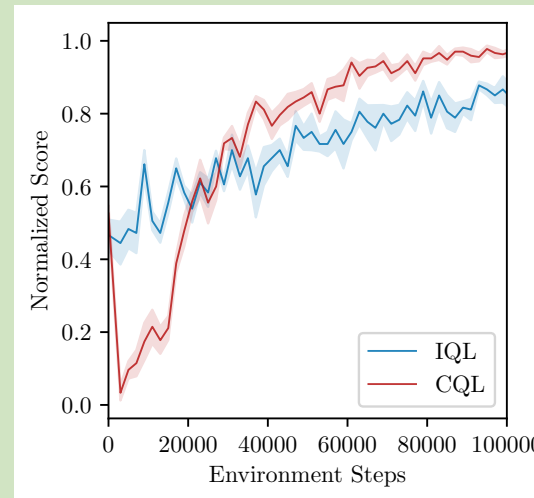
Today's Lecture: Several Facets of Offline RL

Algorithms for Offline RL



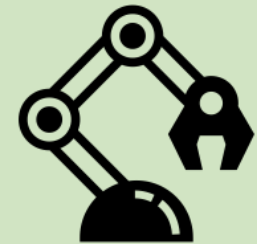
Conservative Q-learning,
Model-based offline RL,
large-scale results

Extensions of Offline RL



Online fine-tuning,
learning with unlabeled data

Applications in Robotics



Robotic pre-training + fine-tuning,
learning from videos

Advanced Topics: Scaling of Offline RL



Train a **single policy** on 40 Atari games

2 billion data-points,
40 games,
sub-optimal data

Lee et al. **Multi-Game Decision Transformers**. NeurIPS 2022 (Oral)

Also related: **Agent 57: Outperforming the Human Atari Benchmark**. DeepMind, 2020.

Kumar et al. **Offline RL on Diverse Multi-Task Data Both Scales and Generalizes**. ICLR 2023 (**Notable top 5%**)

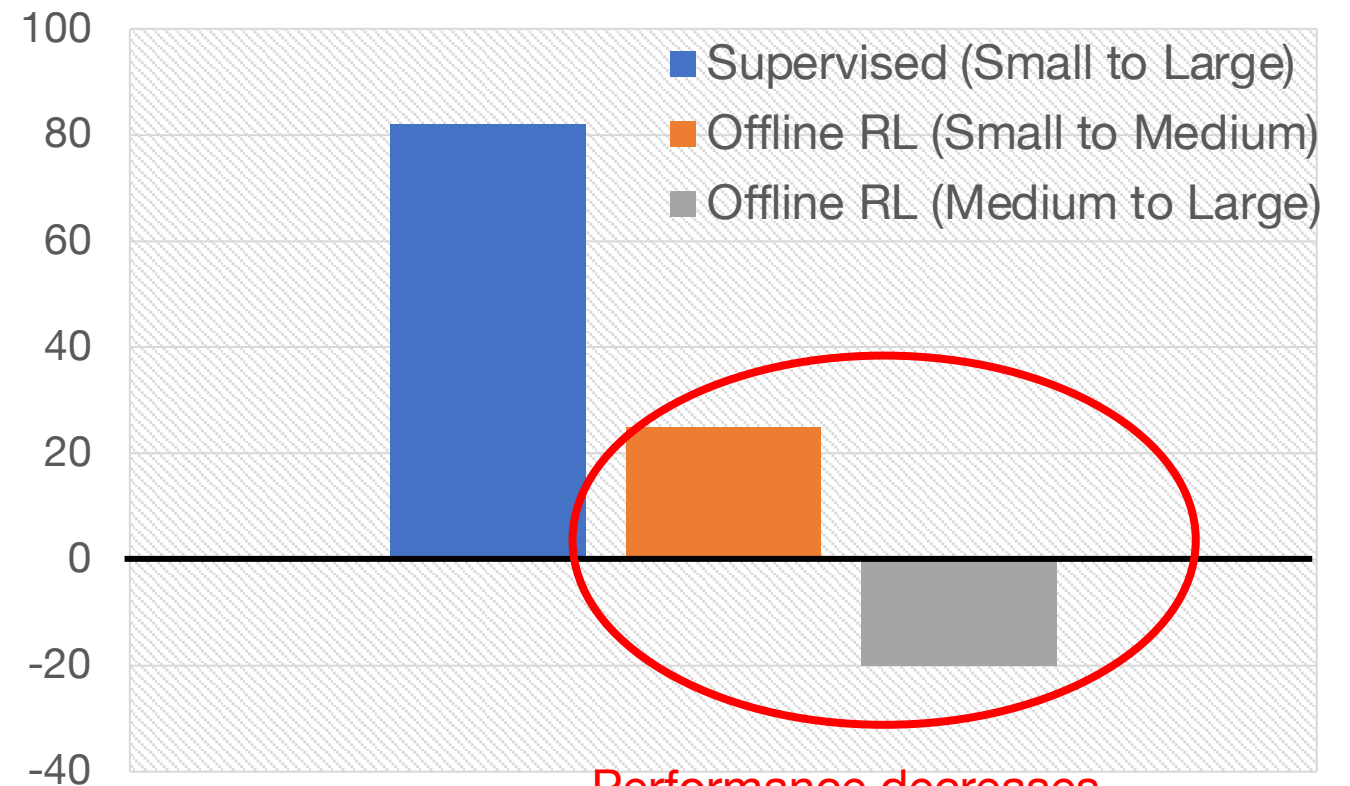
Large-Scale Training of Generalist Policies



Train a **single policy** on 40 Atari games

2 billion data-points,
40 games,
sub-optimal data

**% performance improvement with
model size (Small → Large)**



Performance decreases
with very large models

Why is it hard to use large models in offline RL?

Why do large models behave well in supervised learning?

Implicit regularization

$\min_{\theta} \mathcal{L}(\theta)$

Over-parameterized $\nearrow$

Stochastic gradient descent $\downarrow$

$\min_{\theta} \mathcal{L}(\theta) + \mathcal{R}(\theta)$

$\nwarrow$

“implicit regularizer”

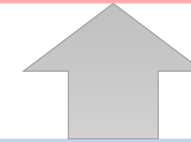
(prefers simpler solutions)

Why is it hard to use large models in offline RL?



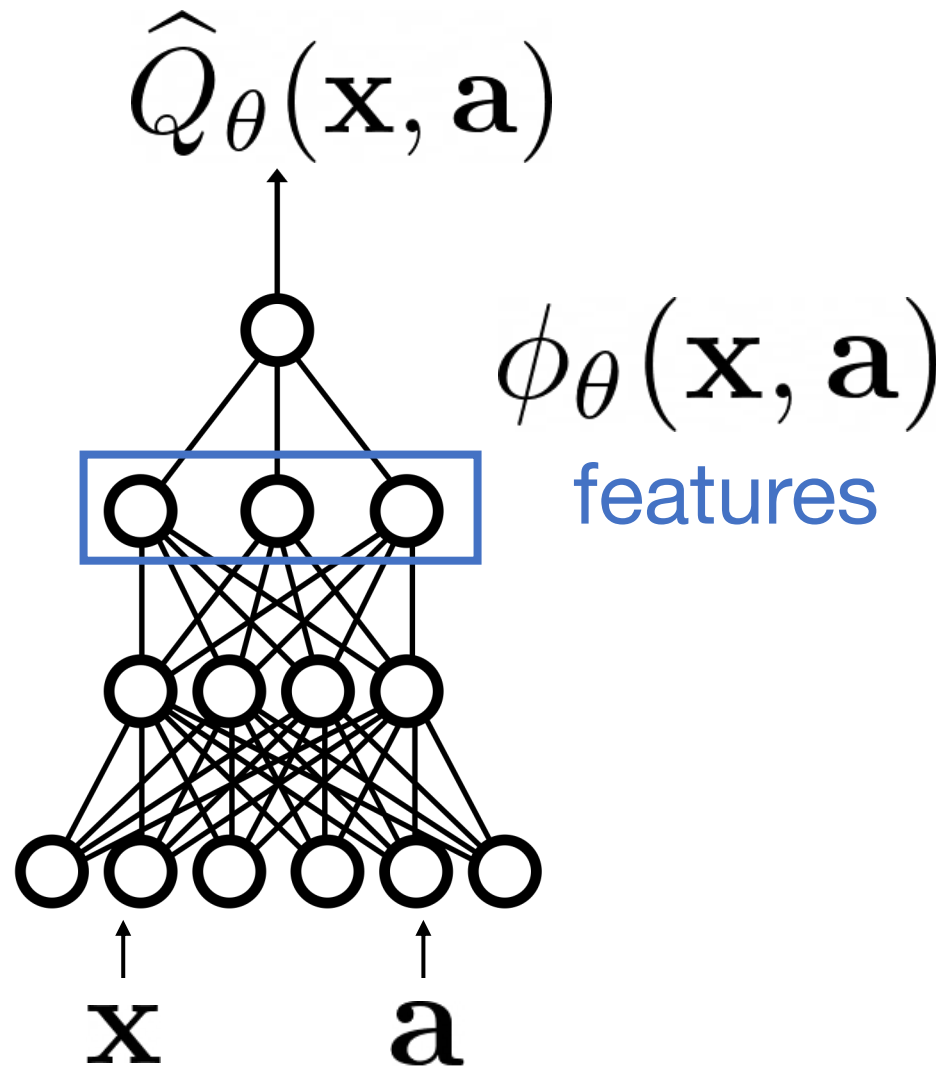
Why do large models behave well in supervised learning?

Implicit regularizer in offline RL??



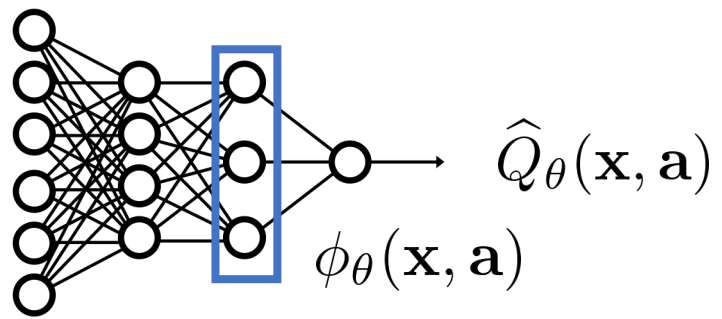
Implicit regularizer prefers “simple” solutions that perform well

Setup: Implicit Regularization in Offline RL



Theory: Implicit Regularization in Offline RL

[Informal] Theorem (Implicit regularization in SL vs offline RL)



Theory: Implicit Regularization in Offline RL

[Informal] Theorem (Implicit regularization in SL vs offline RL)

$$\mathcal{R}_{\text{SL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2$$

Supervised learning (SL)

[Blanc et al. COLT 2021]

$$\mathcal{R}_{\text{RL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2 -$$

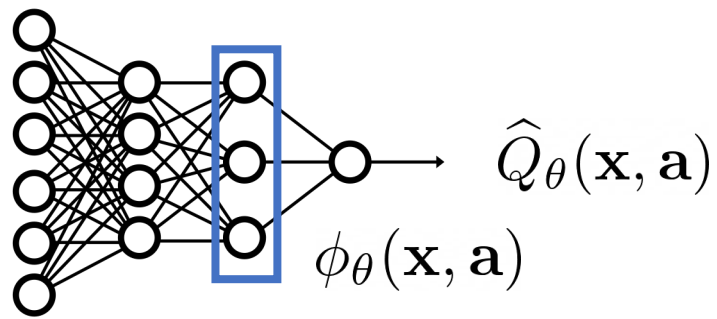
$$\sum_{i=1}^n \mathbb{E}_{\mathbf{a}'_{i+1} \sim \pi} [\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)^{\top} \phi_{\theta}(\mathbf{x}_{i+1}, \mathbf{a}'_{i+1})]$$

Offline RL

[Kumar et al. ICLR 2022]

Reduce feature
lengths / norms

Conflicts by blowing
up feature norms

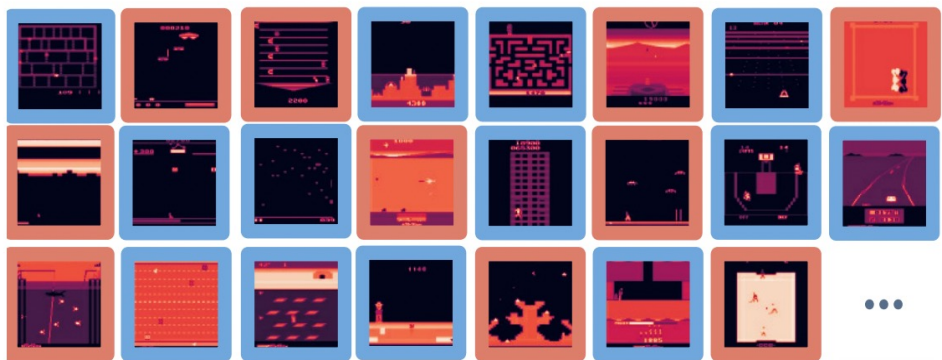


Idea: Can We Undo Implicit Regularization?

$$\mathcal{R}_{\text{SL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2$$
$$\mathcal{R}_{\text{RL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2 - \sum_{i=1}^n \mathbb{E}_{\mathbf{a}'_{i+1} \sim \pi} [\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)^{\top} \phi_{\theta}(\mathbf{x}_{i+1}, \mathbf{a}'_{i+1})]$$

Key Insight: As this term is bad, add it back explicitly during RL

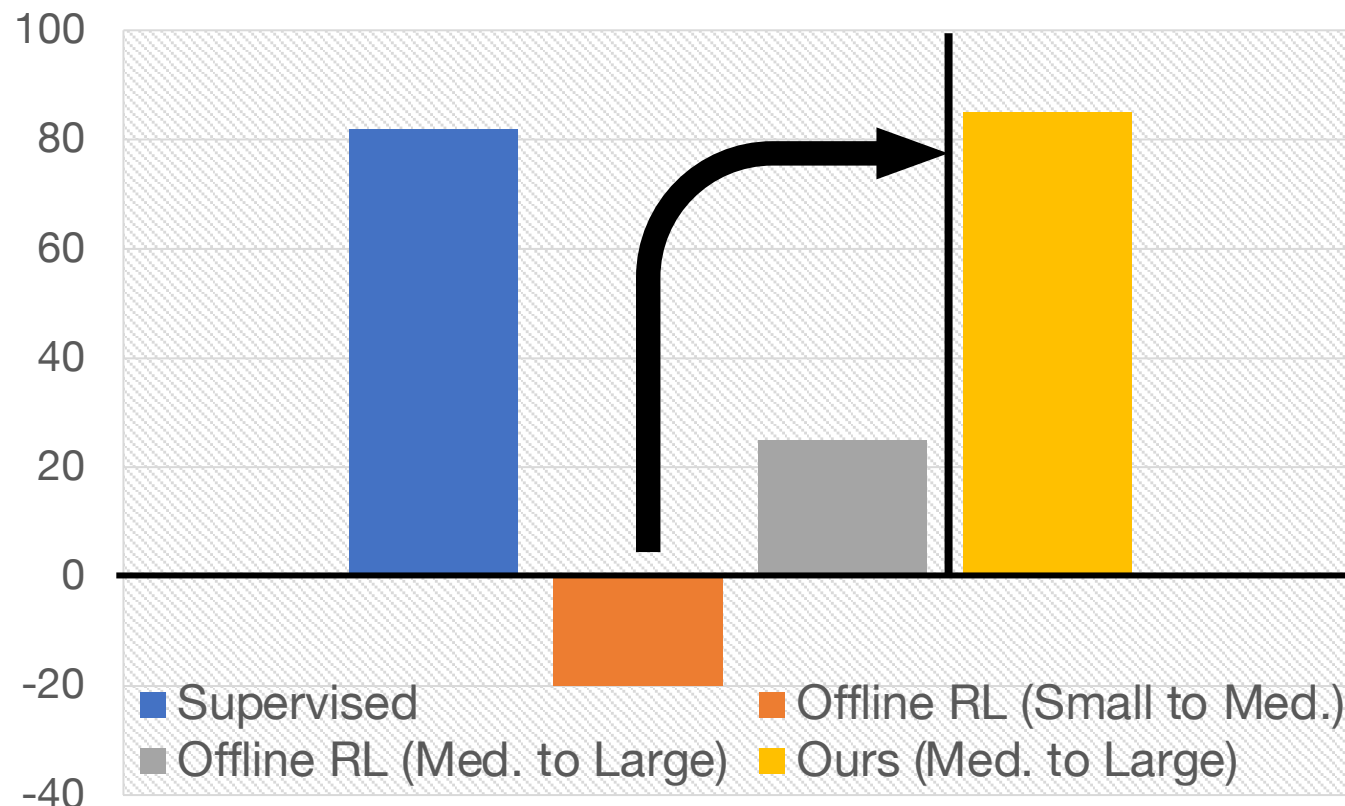
Results: Improves Scalability of Offline RL



Train a **single policy** on 40 Atari games

2 billion data-points,
40 games,
sub-optimal data

**% performance improvement with
model size (Small → Large)**



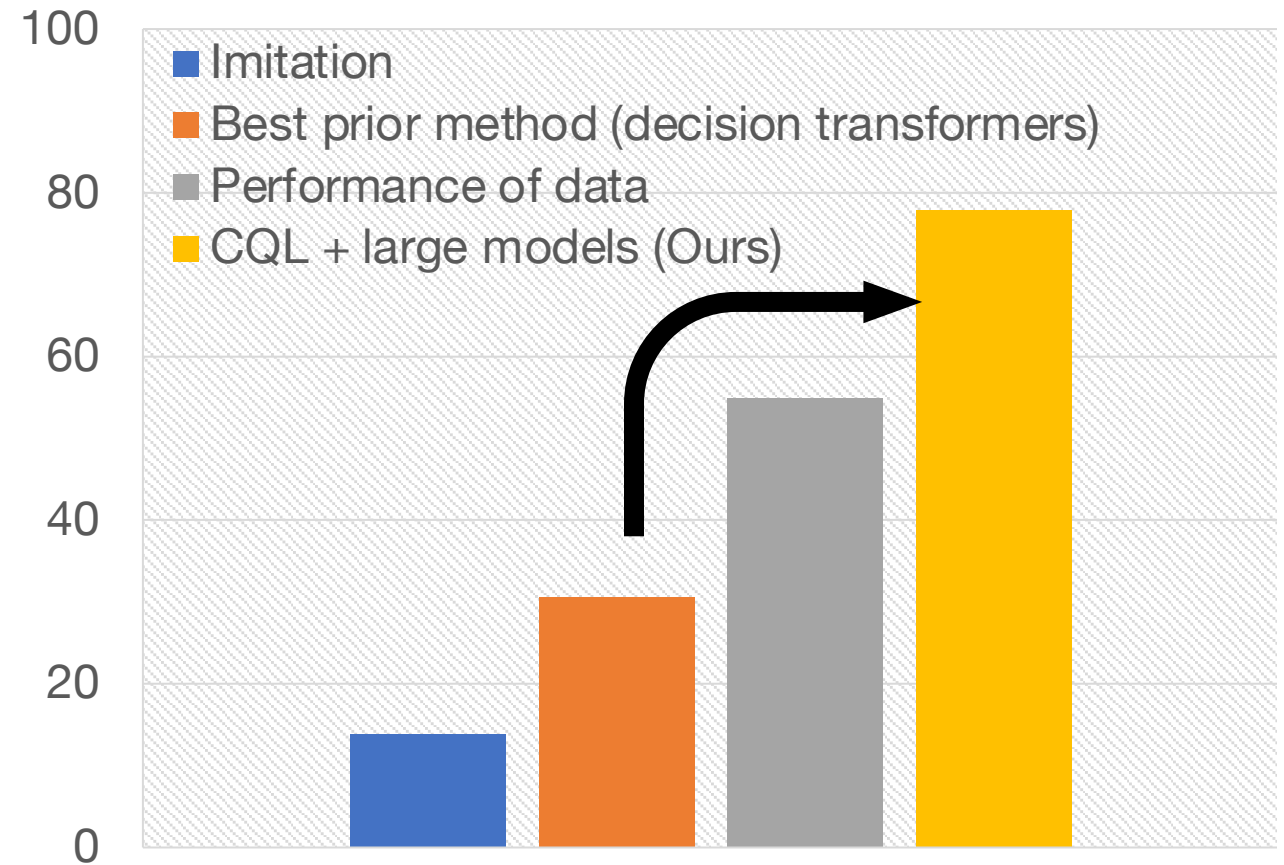
Results: Improves Performance

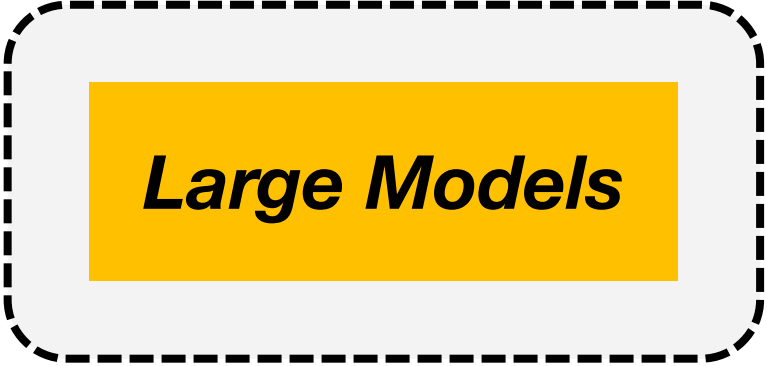


Train a **single policy** on 40 Atari games

First offline RL approach
to train **large models** to
surpass the data!

**Human-normalized score
averaged over 40 games**





Large Models

Takeaways

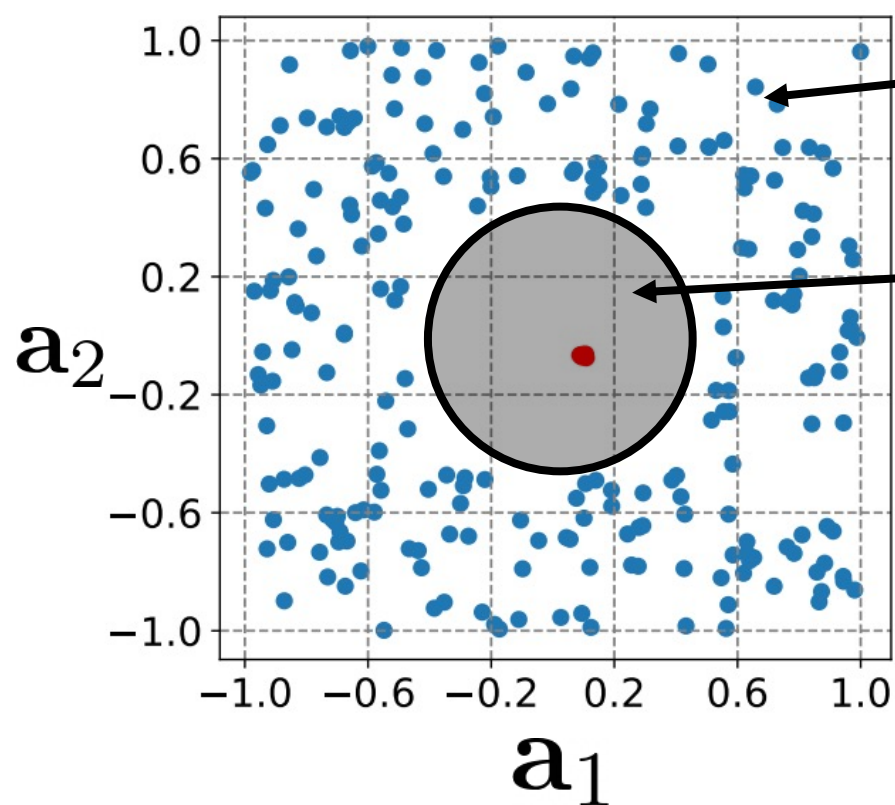
- **Challenge:** Large models can hurt in offline RL
- **Algorithm:** Undoing “bad” part of implicit regularization
- **Result:** Offline RL methods that enjoy benefits of high model capacity in practice

Thank You!

Conservative Offline RL vs Policy Constraint

Intuition: Account for reward function class during offline RL

$$r(\mathbf{a}_1, \mathbf{a}_2) = -(\mathbf{a}_1^2 + \mathbf{a}_2^2)$$



Dataset

Policy constraint methods
will not find these actions

Consider conservative offline RL with
 $\hat{r}_\theta(\mathbf{a}_1, \mathbf{a}_2) = -(f_\theta(\mathbf{a}_1) + g_\theta(\mathbf{a}_2))$

Source of Implicit Regularization in Offline RL

$$\min_{\theta} \mathbb{E}_{\mathbf{x}, \mathbf{a} \sim \mathcal{D}} \left[\hat{Q}_{\theta}(\mathbf{x}, \mathbf{a}) - y(\mathbf{x}, \mathbf{a}) \right]^2$$

Supervised learning

$$y(\mathbf{x}, \mathbf{a}) = r(\mathbf{x}, \mathbf{a})$$

Offline RL

Target depends on the model's predictions

$$y(\mathbf{x}, \mathbf{a}) = r(\mathbf{x}, \mathbf{a}) + \mathbb{E}_{\mathbf{a}'_{i+1} \sim \pi} \left[\hat{Q}_{\theta}(\mathbf{x}_{i+1}, \mathbf{a}'_{i+1}) \right]$$

Intuition: Implicit Regularization in Offline RL

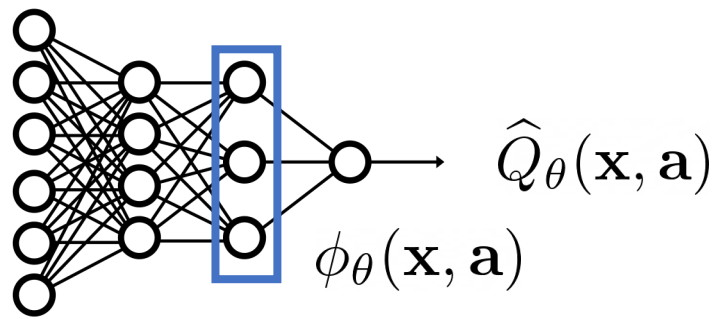
$$\mathcal{R}_{\text{RL}}(\theta) = \sum_{i=1}^n \|\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)\|_2^2 - \sum_{i=1}^n \mathbb{E}_{\mathbf{a}'_{i+1} \sim \pi} [\phi_{\theta}(\mathbf{x}_i, \mathbf{a}_i)^{\top} \phi_{\theta}(\mathbf{x}_{i+1}, \mathbf{a}'_{i+1})]$$

Reduce feature lengths / norms

Offline RL
[Kumar et al. ICLR 2022]

Maximize feature dot products

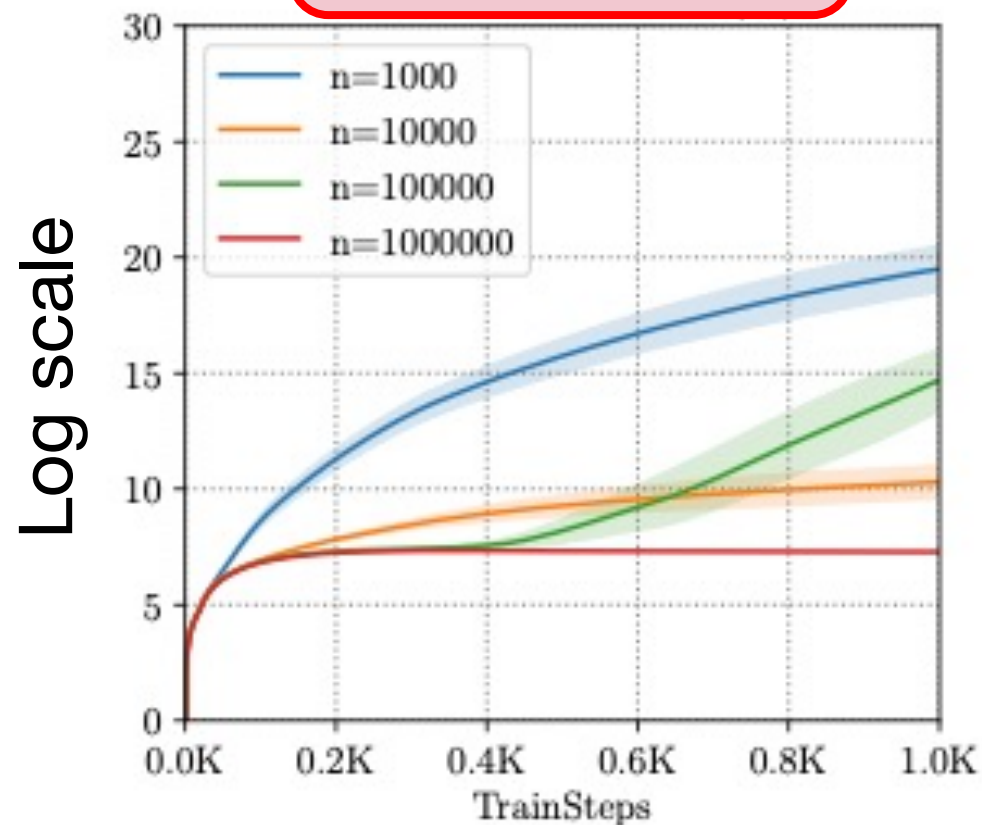
Blows up feature norms



Takeaway: This term has a destabilizing effect

Not Resolved With More Data

(cumulative)
reward model



>>>

Observed reward

